

Virtual MathPsych/ICCM 2024
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A Spectrum of Diffusion Models for Hierarchical Control of Attention: From Sequential to Parallel Processing

Dr. Michael J Frank ~ Brown University
Dr. Matt Nasser ~ Brown University
Dr. Alexander Fengler ~ Brown University
Mr. An Vo ~ Brown University

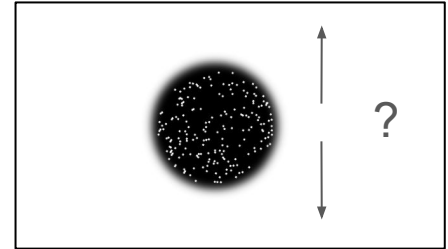
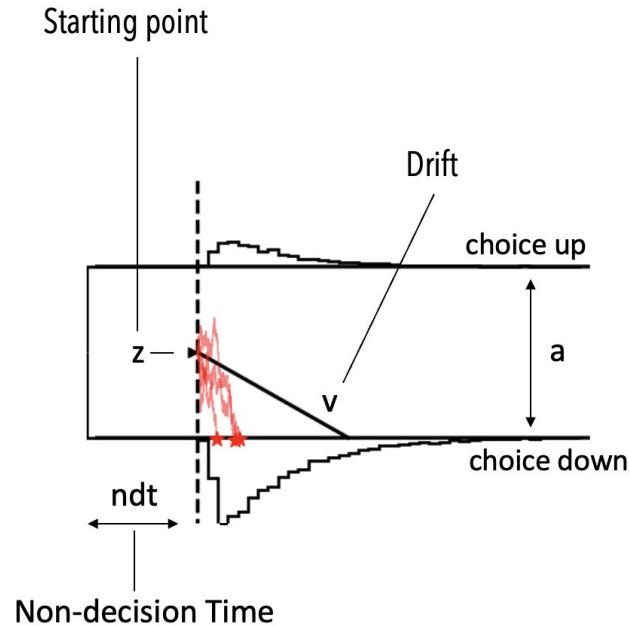


ROBERT J. & NANCY D. CARNEY
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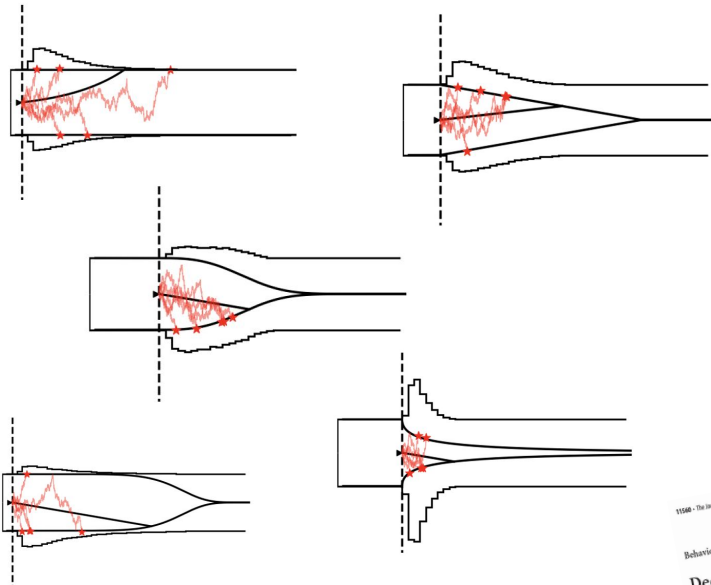
The beloved Drift Diffusion Model...

Very widely applied!



The beloved Drift Diffusion Model...

A host of **variations** on the basic drift diffusion model have been **motivated** in the literature.



Journal List • HHS Author Manuscripts • PMC5450920

HHS Public Access
Author manuscript
Published online 2017 Aug 1
DOI: 10.1016/j.jneurosci.2017.04.008

J. Math. Psychol. Author manuscript; available in PMC 2017 Aug 1.
Published in final edited form as:
J. Math. Psychol. 2018 Aug; 73: 39-79.
Published online 2018 May 24. doi: 10.1016/j.jmp.2018.04.008

Comparing fixed and collapsing boundary versions of the diffusion model
Chaitin Voskuilen,^{a,*} Roger Ratcliff,^a and Philip L. Smith^b

* Author information • Copyright and License information • Disclaimer

11008 • The Journal of Neuroscience, September 26, 2008 • 28(39):11008–11011

Behavioral/Systems/Cognitive

Decisions in Changing Conditions: The Urgency Model

Paul Clark, Genevieve Aude Poulton, and Steph
Gauthier de Boerhaere sur le Système Nerveux Central, Uqam

Several widely accepted models of decision making
is reached and a decision is made. These models are
by neurophysiological studies showing that neurons
to task difficulty and reaches a relatively constant

Jumping to Conclusion?
A Lévy Flight Model of Decision Making

Eva Marie Wieschen,^{a,1} Andreas Voss,^{a,1} & Stefan Rader^{a,2}

^aHamburg University

Psychonomic Bulletin & Review 2018 25:613-632
https://doi.org/10.3758/s13428-018-1060-4

THEORETICAL REVIEW

Sequential sampling models with variable boundaries and non-normal
noise: A comparison of six models

Brian Voss,¹ Veronika Lerche,¹ Ulf Merten,¹ Jochen Voss¹

Received: 16 January 2019
1 Author(s) 2019

KEY
The most prominent response-time models in cognitive psychology is the diffusion model, which assumes that decisions
is based on a continuous evidence accumulation described by a Wiener diffusion process. In the present paper, we
two basic assumptions of standard diffusion model analyses. Firstly, we address the question of whether participants

Journal List • J. Neurosci. • 2015 Feb 11 • PMC6605613

J. Neurosci., 2015 Feb 11; 35(6): 2476–2484.
doi: 10.1523/JNEUROSCI.2410-14.2015

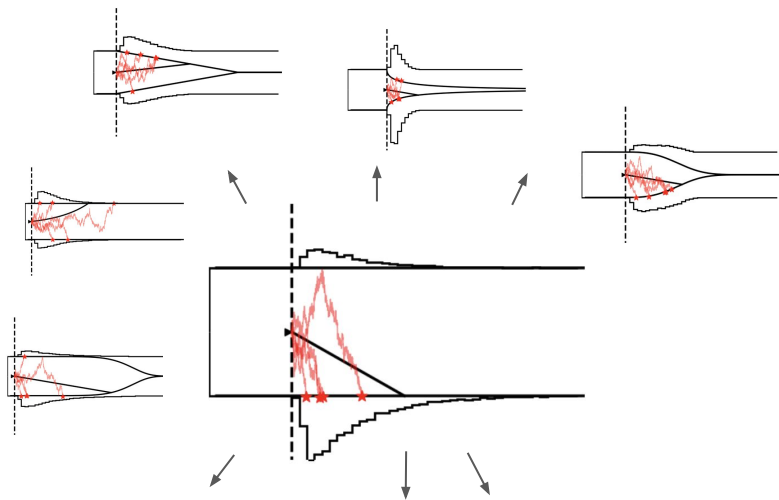
Revisiting the Evidence for Collapsing Boundaries and Urgency Signals in Perceptual
Decision-Making

Gay E. Hawkins,¹ Birte U. Forstmann,² Eric-Jan Wagenmakers,³ Roger Ratcliff,⁴ and Scott D. Brown¹

* Author information • Article notes • Copyright and License information • Disclaimer

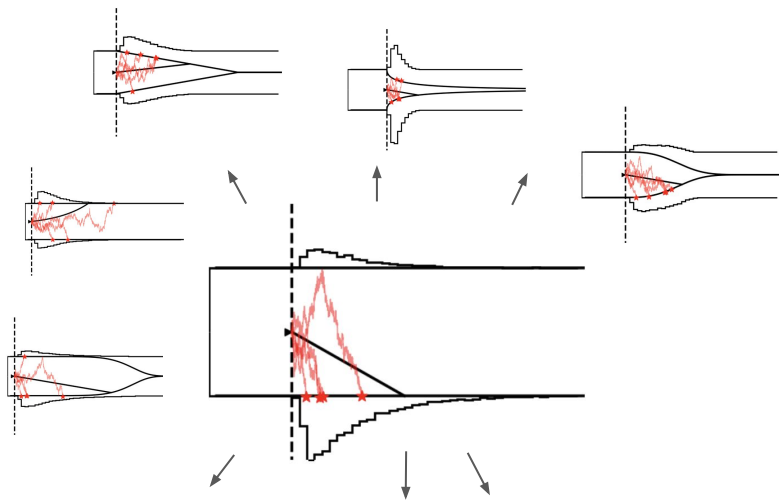
PMC6605613
PMID: 25673842

Outsized influence of DDM



Until recently **empirical data analysis** has put an **outsized focus** on the basic **DDM**.

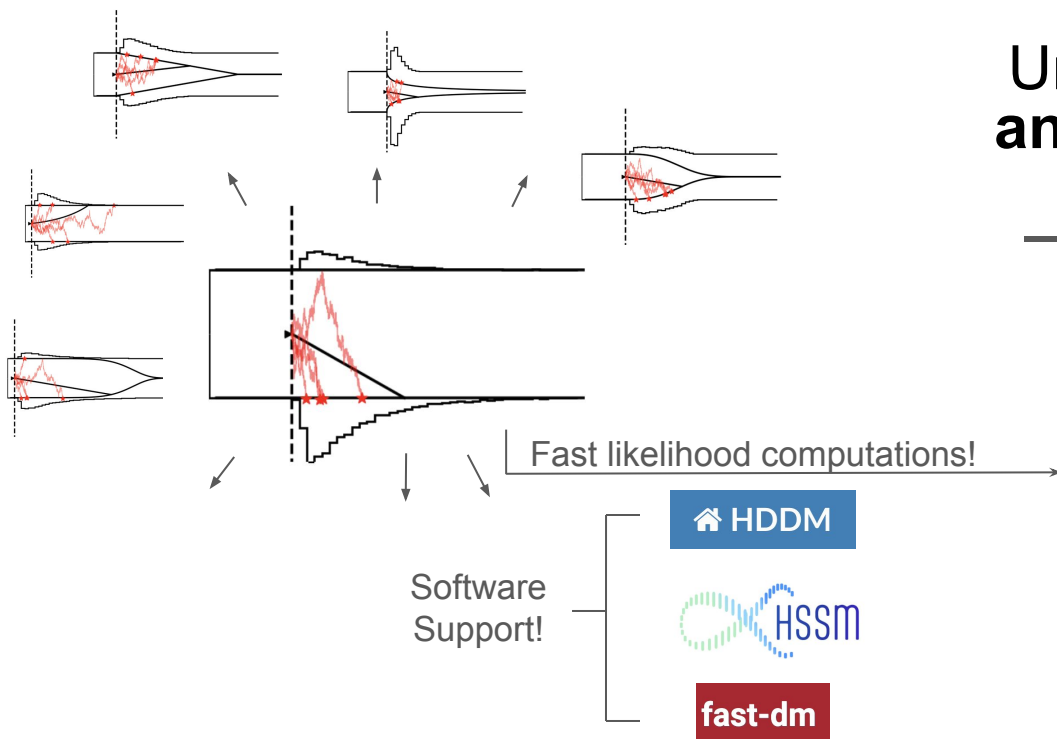
Outsized influence of DDM



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Why?

Outsized influence of DDM



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Why?

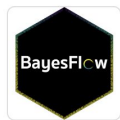
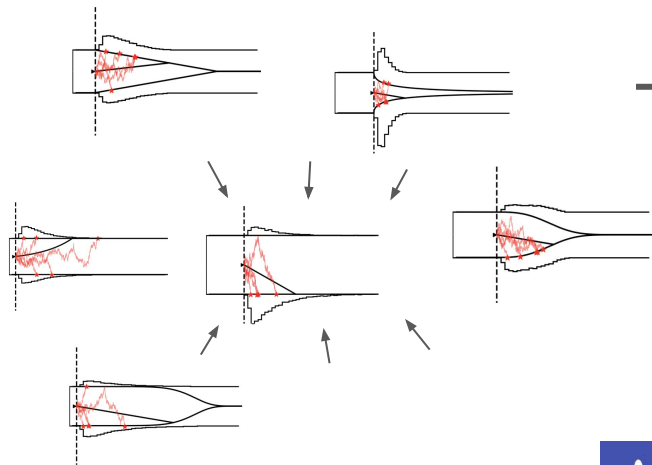
$$p_M(rt, c = 1|v, a, z) = \frac{\pi}{a^2} \exp(-vaz - \frac{v^2 t}{2}) \sum_{k=1}^{\infty} k \exp\left(-\frac{k^2 \pi^2 t}{2a^2}\right) \sin(k\pi z)$$

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

SBI / New Toolkit

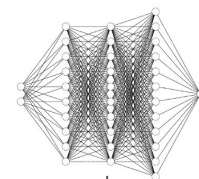
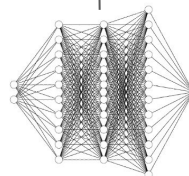
Recent **boost** in methods and software support, **leveled the playing field!**

How?

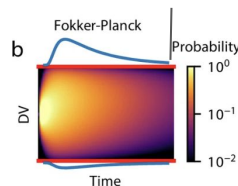


BayesFlow
SNPE
...

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$



LANs
MNLEs
SNLE
...

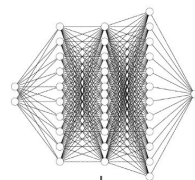
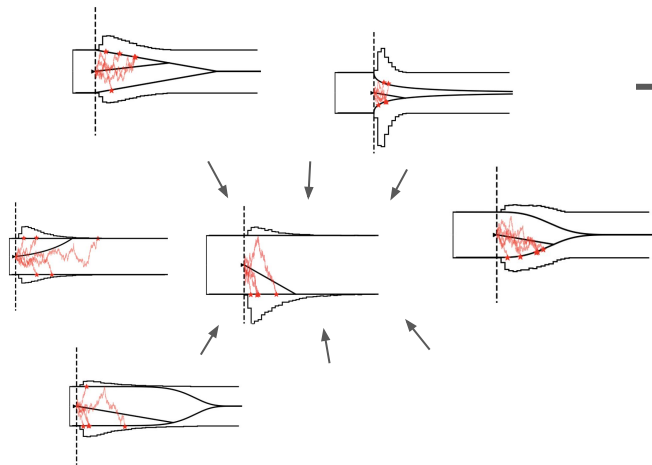


PyDDM



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How?



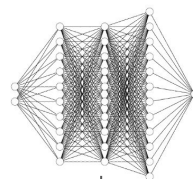
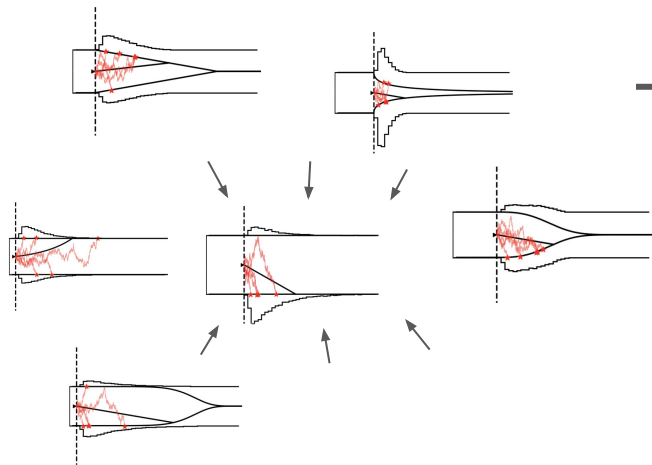
$$p(\theta|x) \propto p(x|\theta)p(\theta)$$



LANs
MNLEs
SNLE
...

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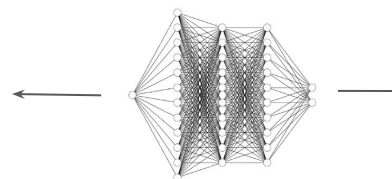
LANs
MNLEs
SNLE
...

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

Key Idea:

Learn from simulations once → reuse

Small ad...



LANs
MNLEs
SNLE

...

<https://Inccbrown.github.io/HSSM/>

[Home](#) [Getting Started](#) [API References](#) [Tutorials](#) [Contributing](#)

[HOME](#)
[Overview](#)
[Credits](#)
[License](#)
[Changelog](#)

Overview



pypi v0.2.1

downloads 511/month

python 3.10 | 3.11

pull requests 5 open

build passing

Stars 68

code style black

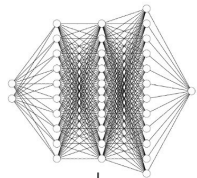
HSSM (Hierarchical Sequential Sampling Modeling) is a modern Python toolbox that provides state-of-the-art likelihood approximation methods within the Python Bayesian ecosystem. It facilitates hierarchical model building and inference via fast and robust MCMC samplers. User-friendly, extensible, and flexible, HSSM can rigorously estimate the impact of neural and other trial-by-trial covariates through parameter-wise mixed-effects models for a large variety of cognitive process models.

HSSM is a [BRAINSTORM](#) project in collaboration with the [Center for Computation and Visualization \(CCV\)](#) and the [Center for Computational Brain Science](#) within the [Carney Institute at Brown University](#).

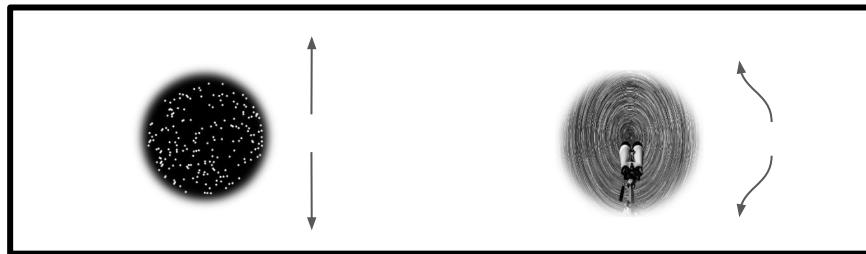
New horizons...



These new methods allow us to become more **ambitious** with our **computational models**, and in turn about respective **experimental designs**!



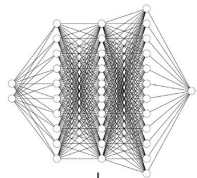
$$p(\theta|x) \propto p(x|\theta)p(\theta)$$



New horizons...

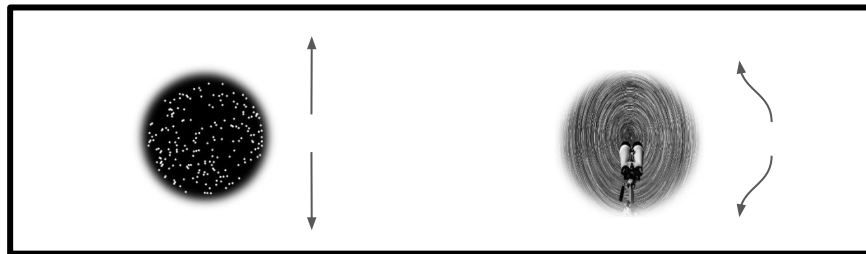


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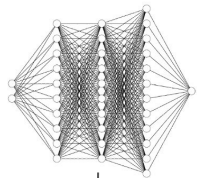
Re-analysis of complicated experiments via computational modeling



New horizons...



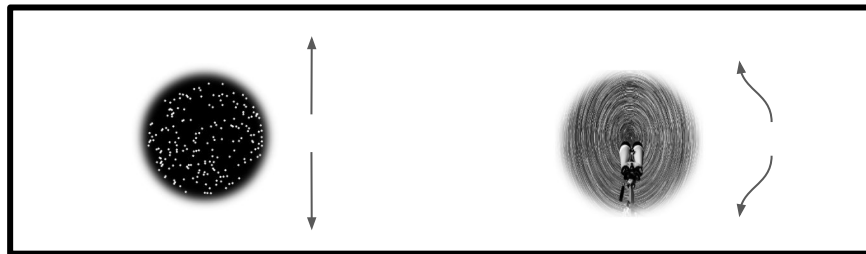
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Re-analysis of complicated experiments via computational modeling

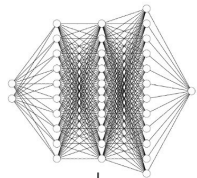
Influence the design of ***new experiments***



New horizons...



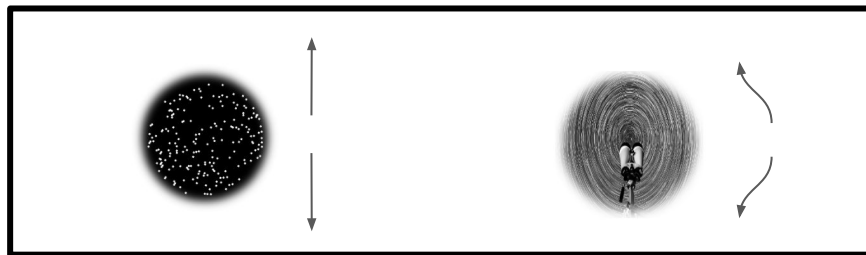
These new methods allow us to become more **ambitious** with our **computational models**, and in turn about respective **experimental designs**!



$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

Re-analysis or
complicated experiments
via computational
modeling

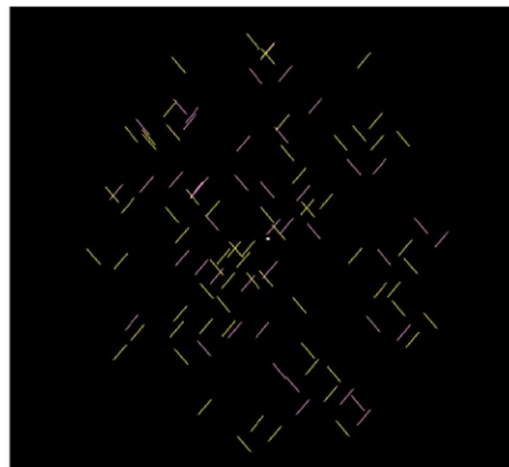
Influence the design of
new experiments



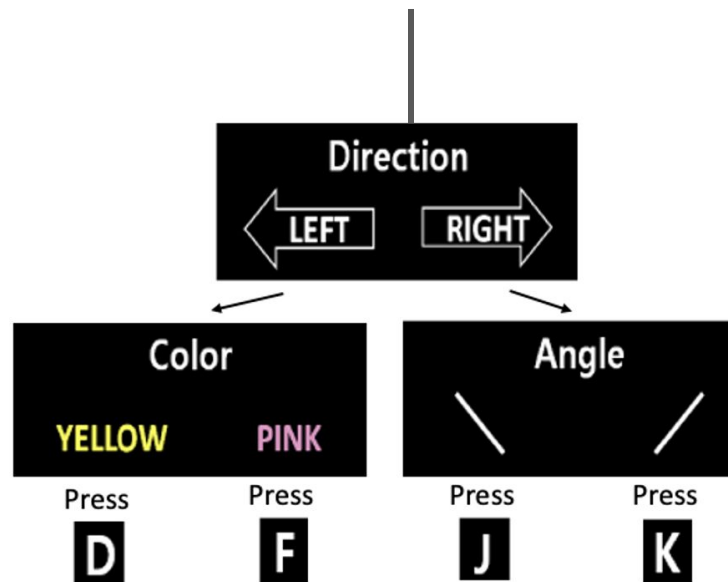
Setting...



Three features



Stimulus Screen

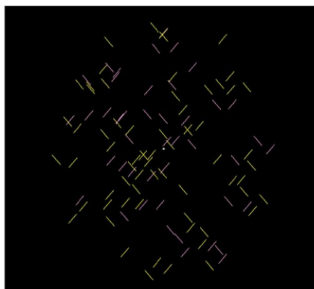


Hierarchical
Task Design

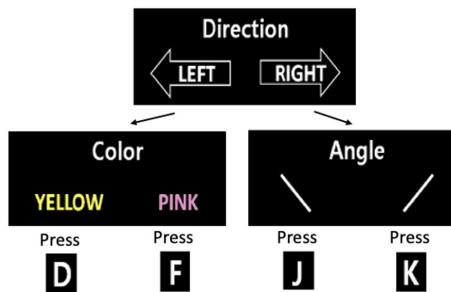
Task Instructions

4 choice options

Hierarchical Decision Making



Stimulus Screen



Task Instructions

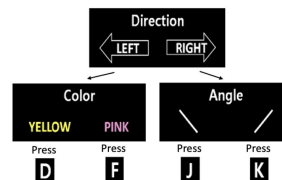
Overarching question:

Does the *hierarchical task structure* induce a *sequential or parallel decision making* process?

Hierarchical Decision Making



Stimulus Screen



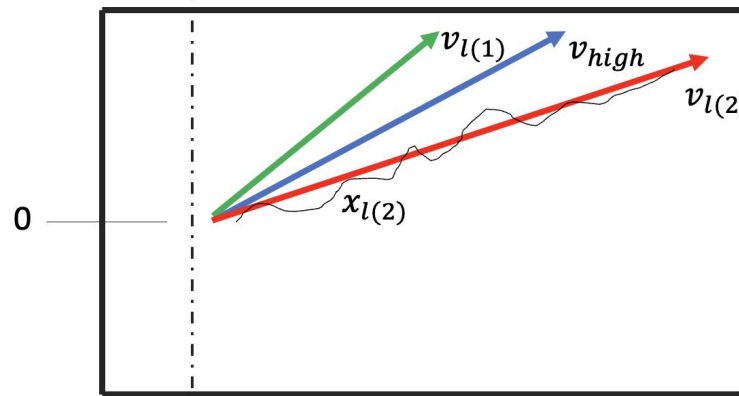
Task Instructions

Overarching question:

Does the *hierarchical task structure* induce a *sequential or parallel decision making* process?

The goal is to test three types of cognitive process models:

1. **Parallel Processing**
2. **Sequential Processing**
3. **Resource Rational Hybrids**



choice is triggered after
both high dim and
relevant low dim
processes are finished

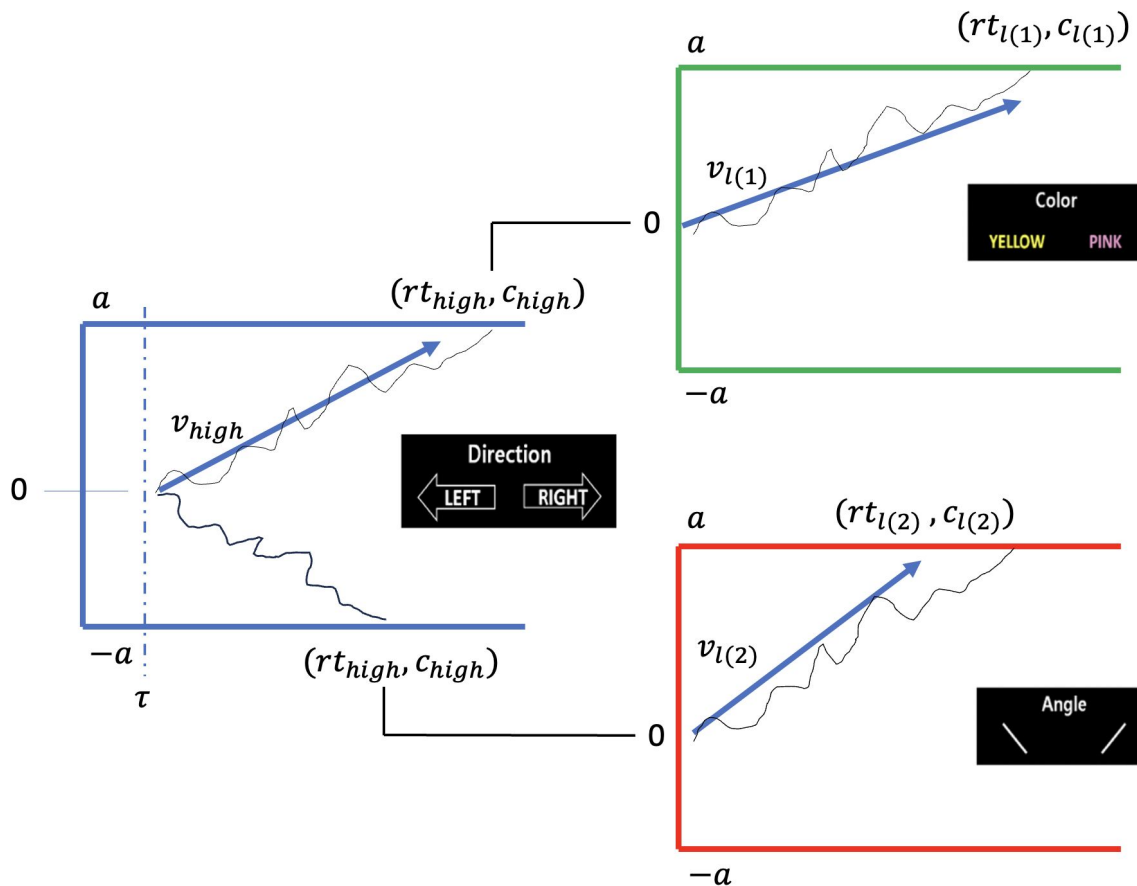
$$rt = \max(rt_{high}, rt_{l(c_h)})$$

$$c = (c_h, c_{l(c_h)})$$

high dim choice

relevant low dim choice

Sequential Processing



RT is the simple **sum**
of **high dim** and
relevant low dim
process

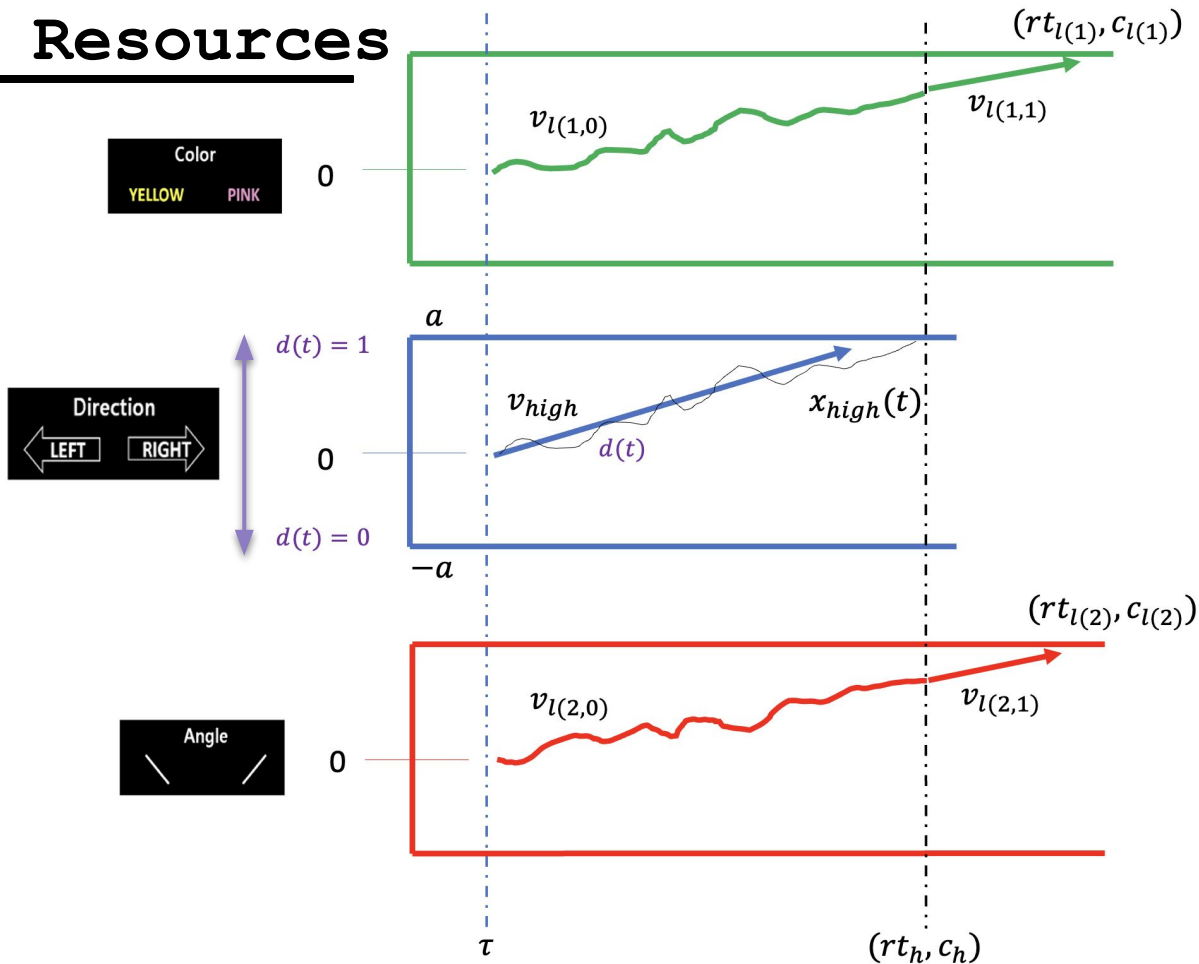
$$rt = rt_{high} + rt_{l(c_{high})}$$

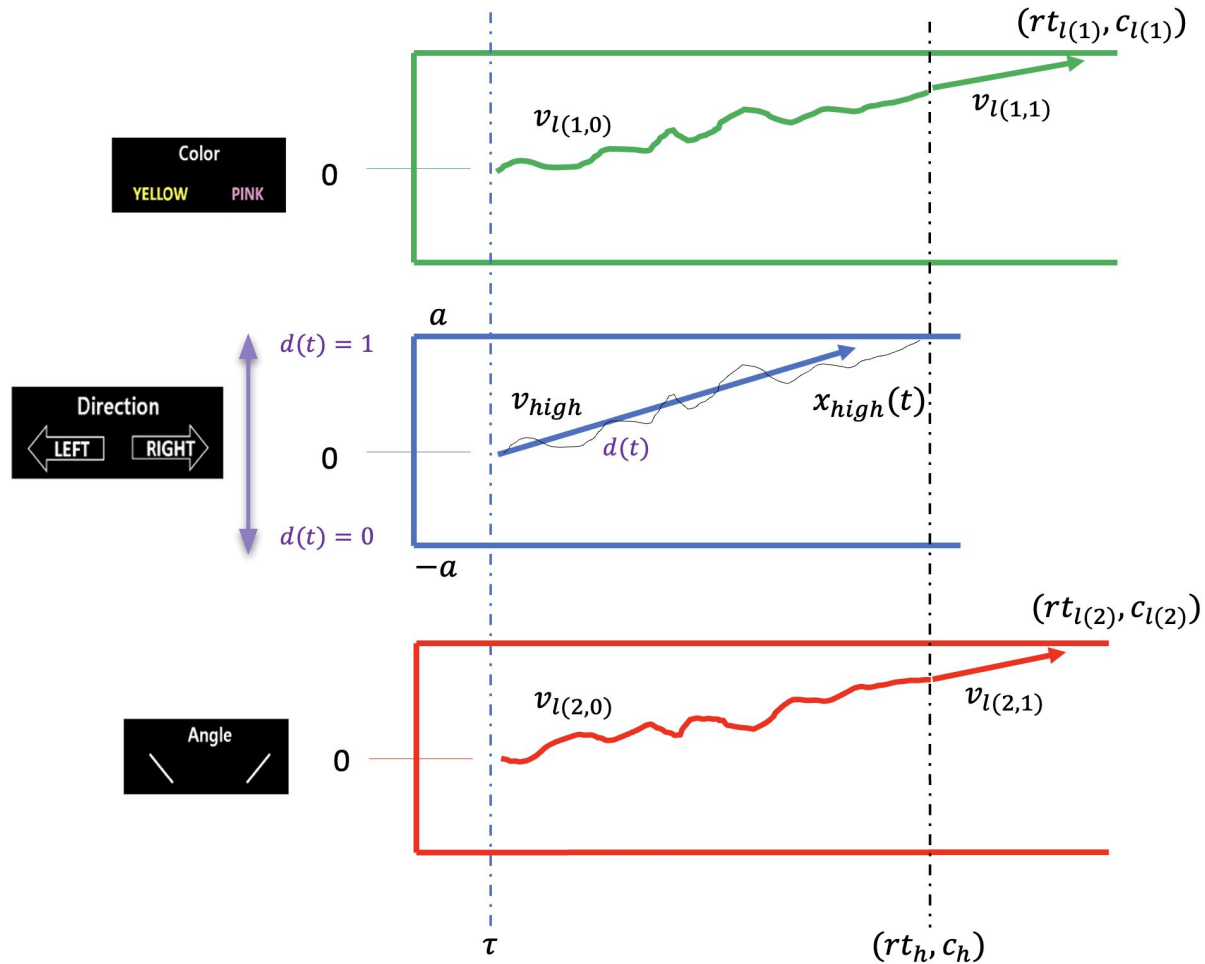
$$c = (c_{high}, c_{l(c_{high})})$$

high dim choice

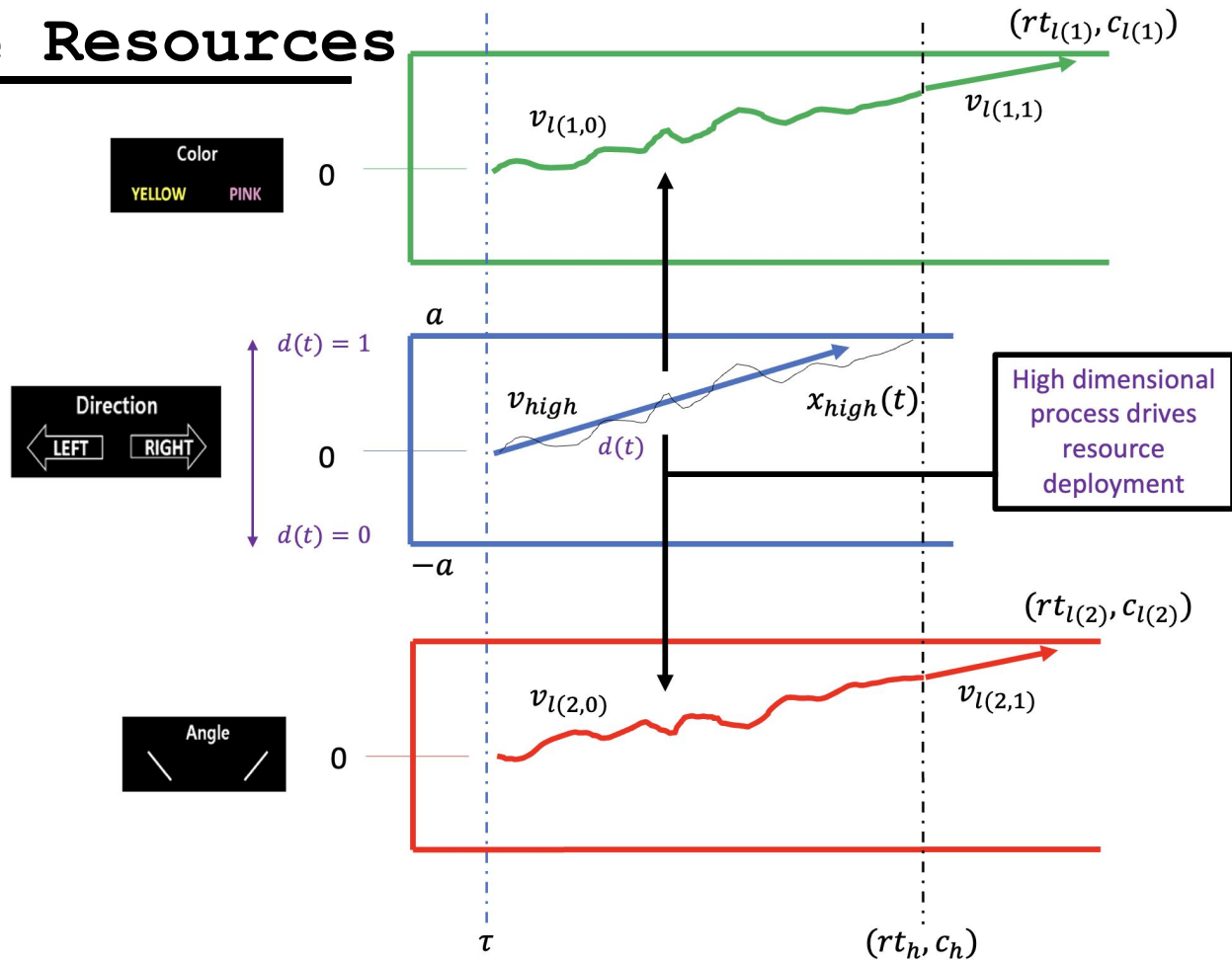
relevant low dim choice

Hybrid: Finite Resources





Hybrid: Finite Resources



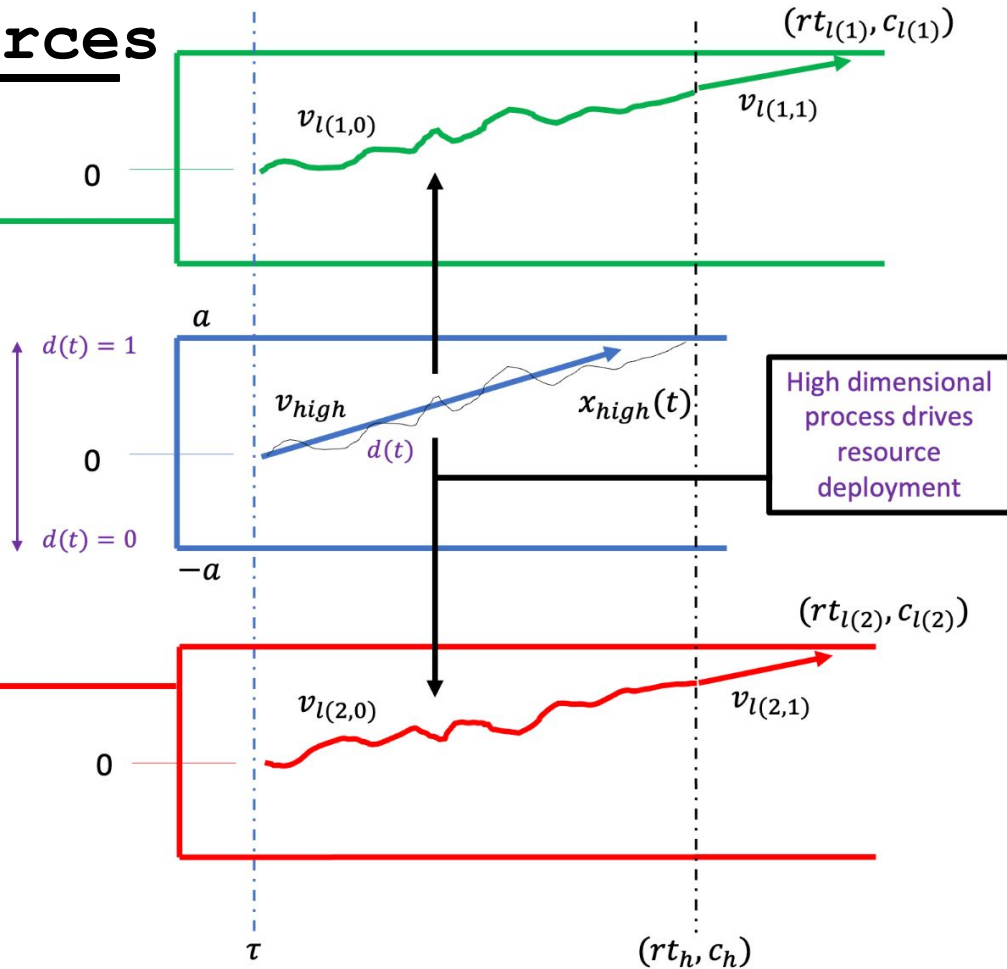
Hybrid: Finite Resources

$$v_{l(1,0)}(t) = (1 - d) * d(t) * v_{l1}$$

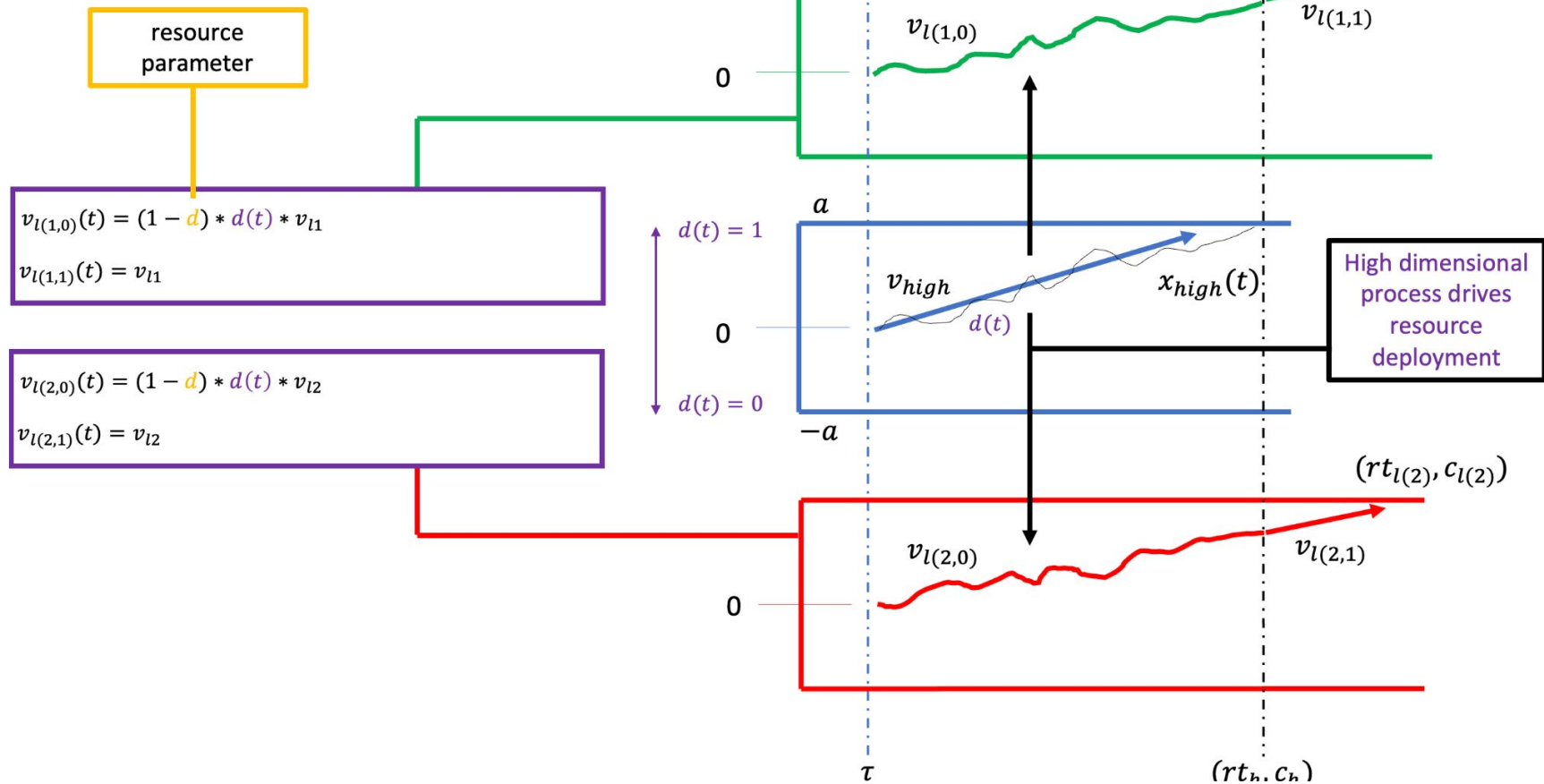
$$v_{l(1,1)}(t) = v_{l1}$$

$$v_{l(2,0)}(t) = (1 - d) * d(t) * v_{l2}$$

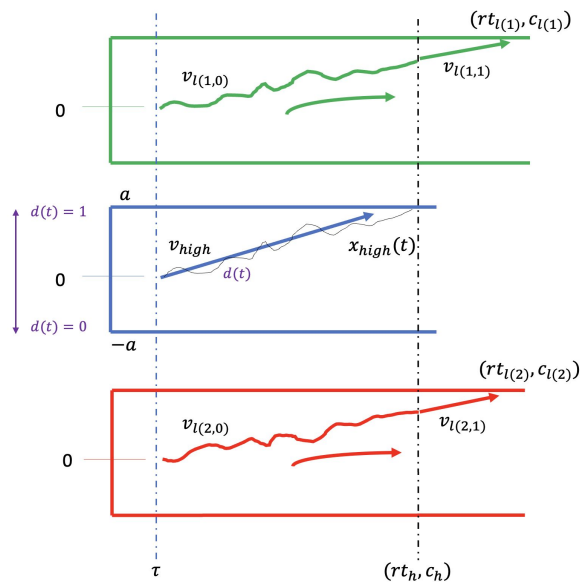
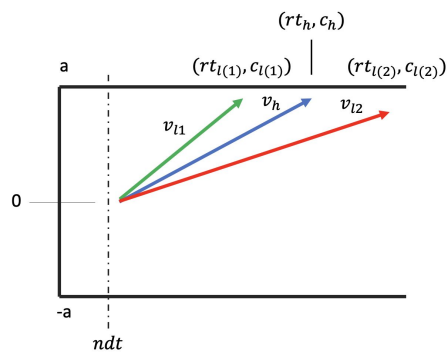
$$v_{l(2,1)}(t) = v_{l2}$$



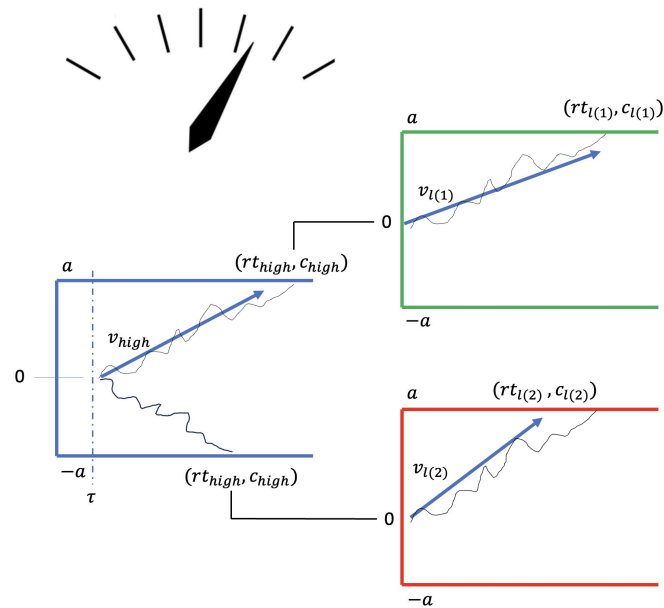
Hybrid: Finite Resources



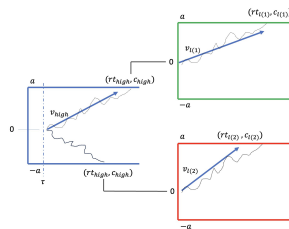
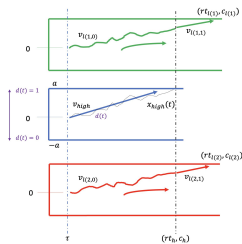
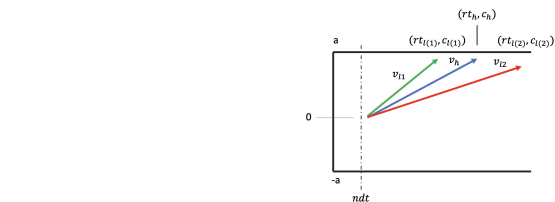
Fully Parallel



Fully Sequential

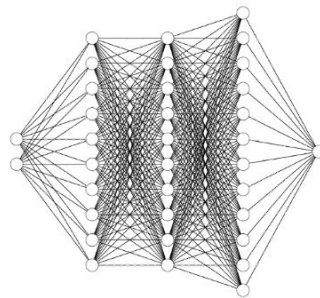


Strategy



Forward Simulate
(Trivial)

Train LANs



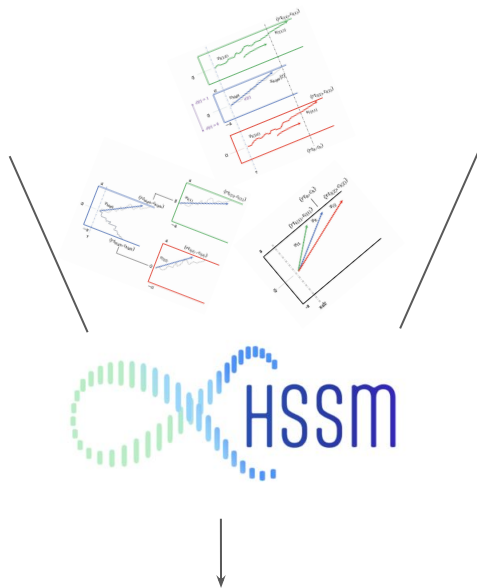
Toolbox



$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

- Hierarchies
- Trial-wise neural regressors by parameter etc.

Strategy

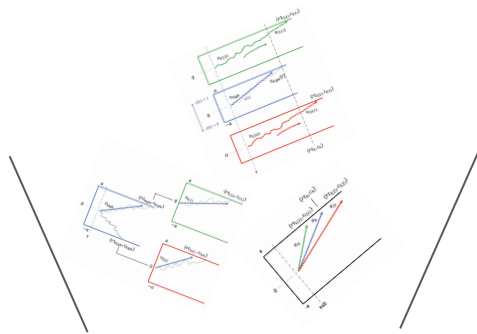


- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

We have the networks in the toolbox,
what next?

1. **Parameter recovery** studies on synthetic data mimicking structure of empirical dataset
 - a. Simple fits
 - b. Hierarchical models

Strategy

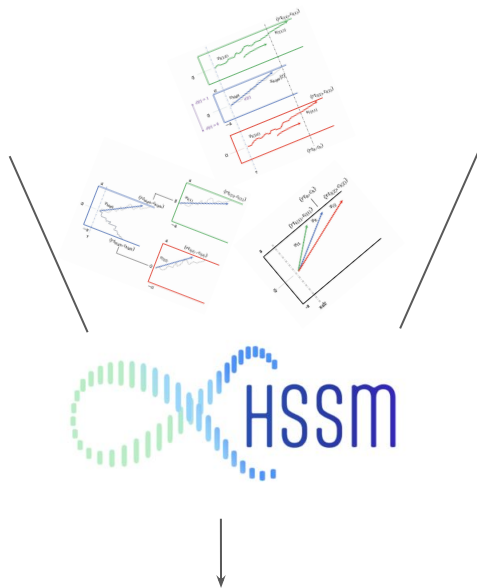


- Hierarchies
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- etc.

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 - a. Simple fits
 - b. Hierarchical models
2. Analysis on **empirical data**
 - a. Simple fits
 - b. Hierarchical models
 - c. Include trial-wise neural regressors (introduce EEG covariates)

Strategy

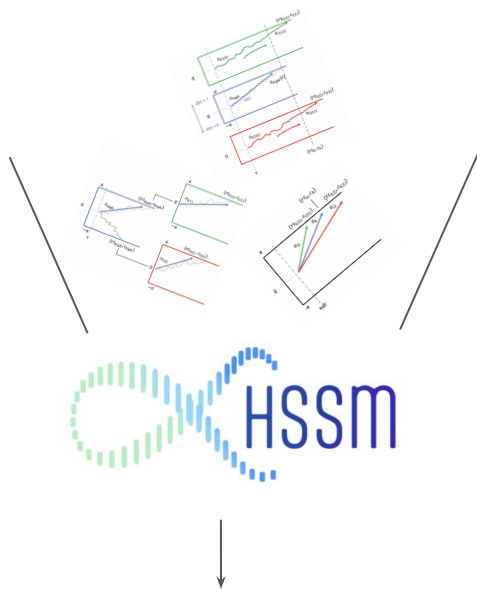


- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

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3. **Model Comparison**

Strategy

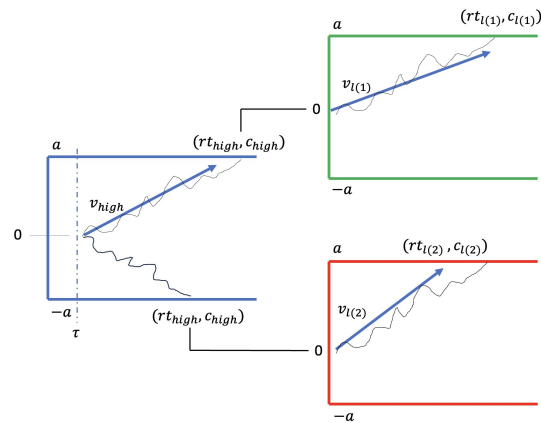
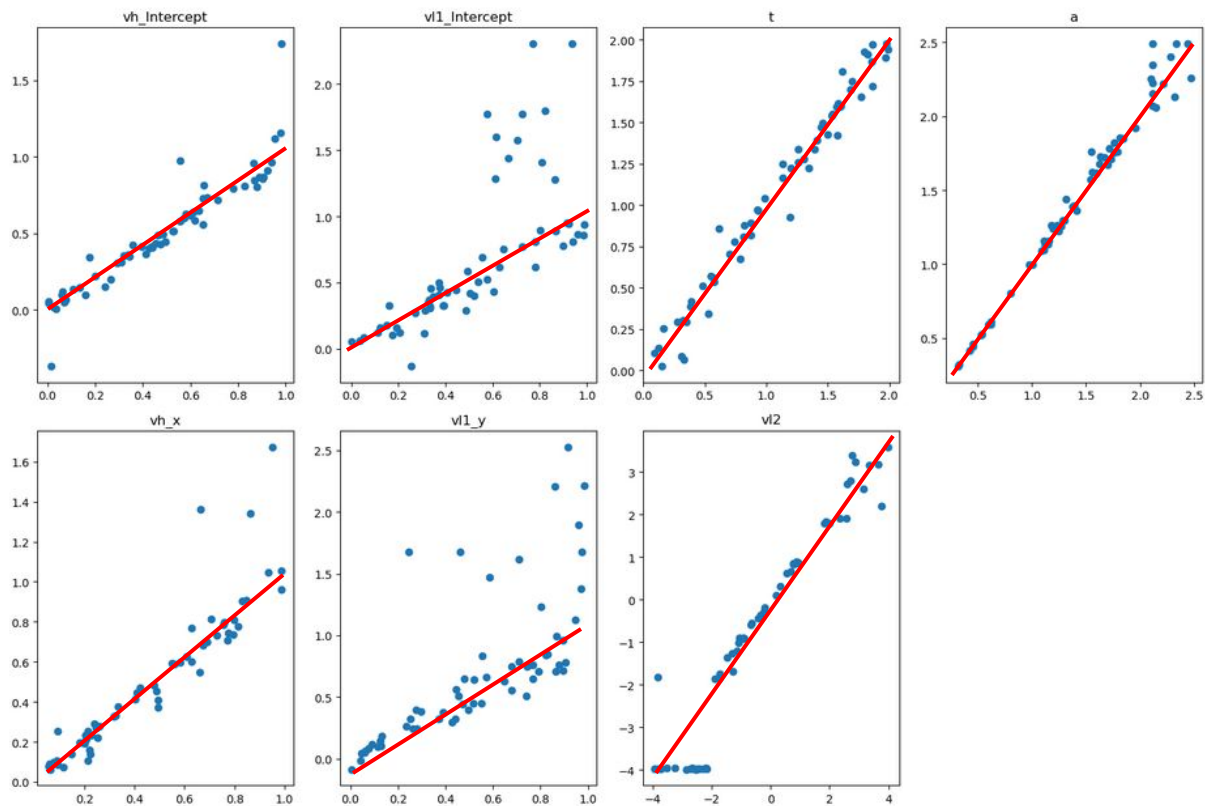


- Hierarchies
- Trial-wise neural regressors by parameter
- etc.

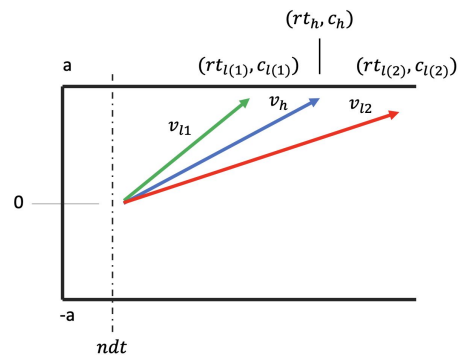
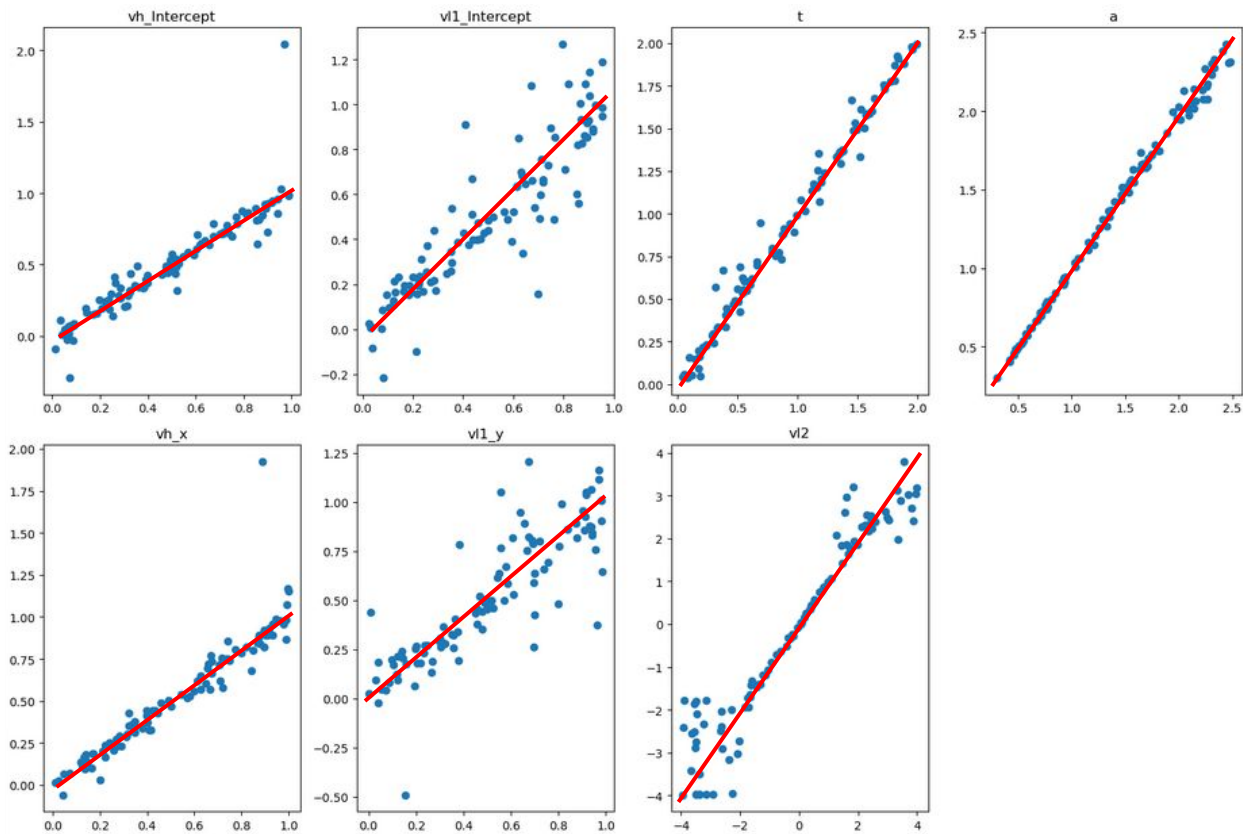
We have the networks in the toolbox, what next?

1. **Parameter recovery** studies on synthetic data mimicking structure of empirical dataset
 - a. Simple fits  We are here
 - b. Hierarchical models
2. Analysis on **empirical data**
 - a. Simple fits
 - b. Hierarchical models
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3. **Model Comparison**

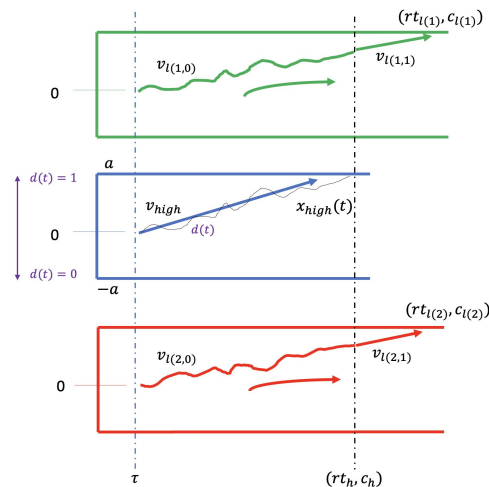
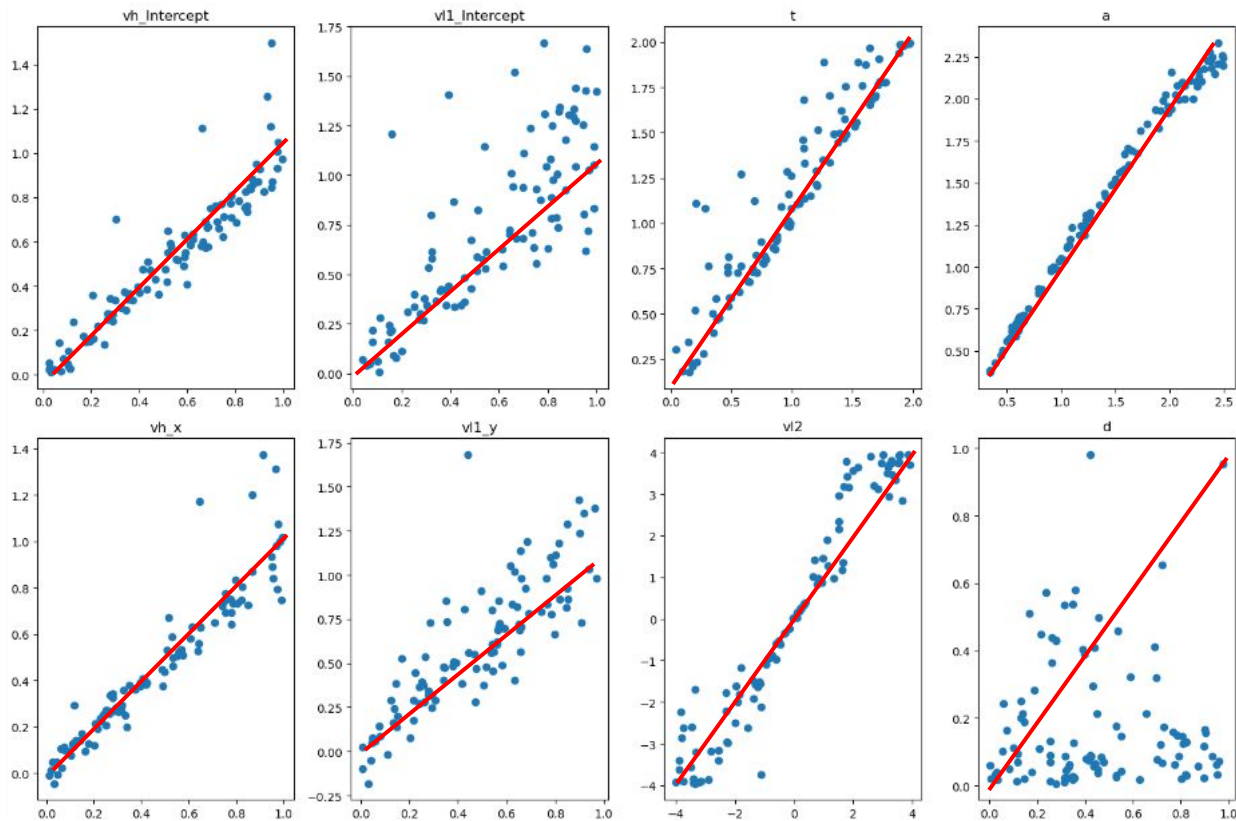
Parameter Recovery: Sequential



Parameter Recovery: Parallel



Parameter Recovery: Hybrid



Trouble
identifying d
parameter

End



Michael Frank



An Vo



Frank Lab



Paul Xu



Matt Nasser



ROBERT J. & NANCY D. CARNEY
INSTITUTE FOR BRAIN SCIENCE
BROWN UNIVERSITY

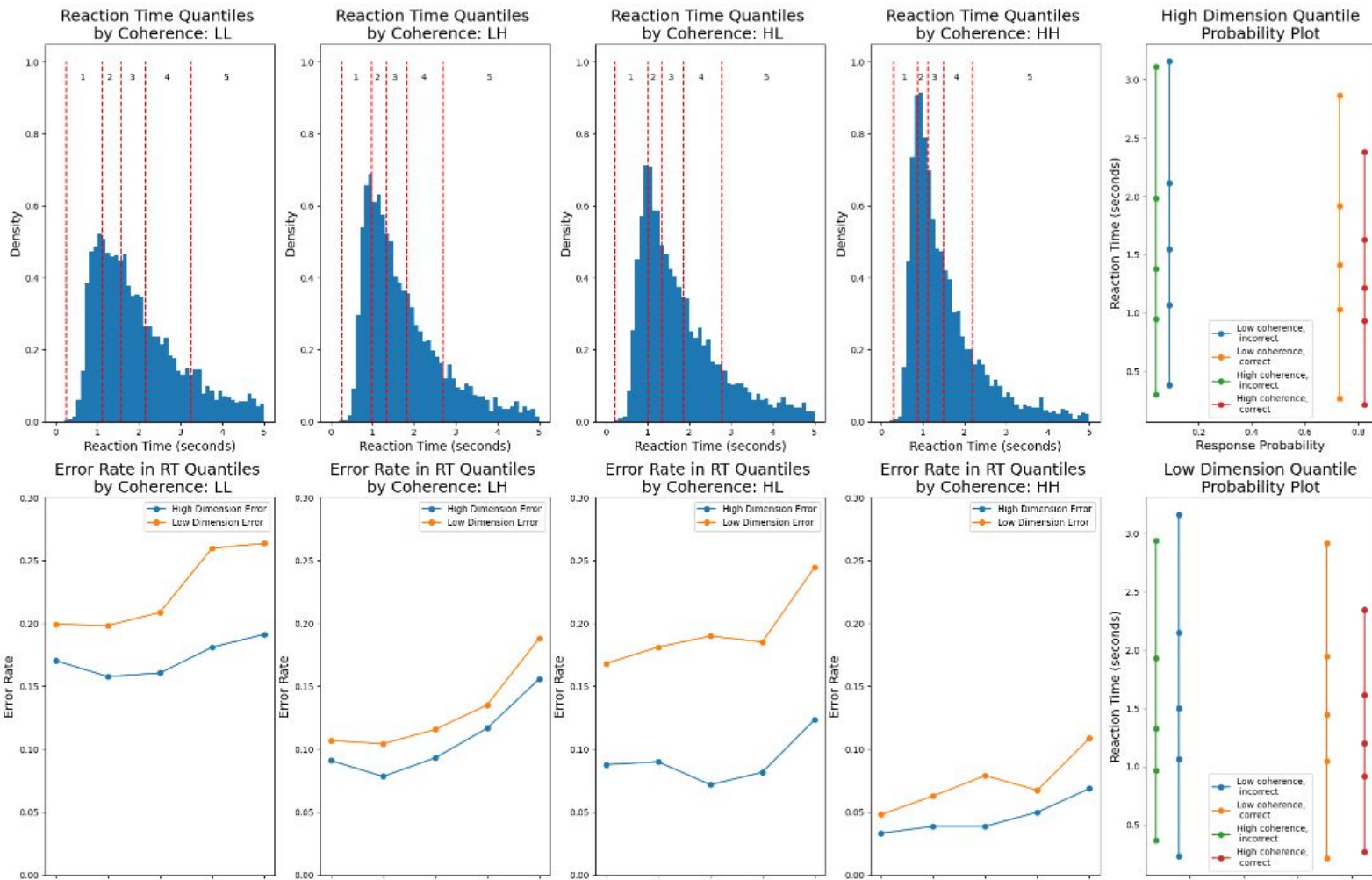


References

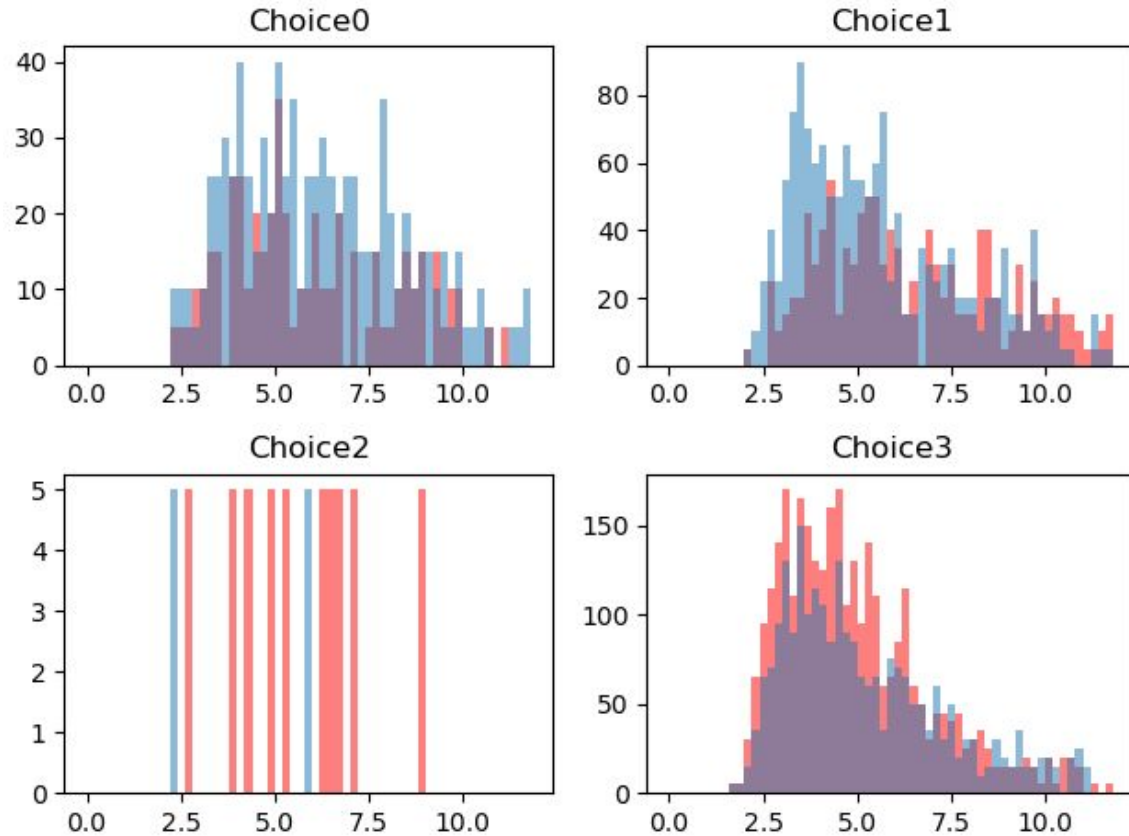
1. Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion decision model: Current issues and history. *Trends in cognitive sciences*, 20(4), 260-281.
2. Tejero-Cantero, Alvaro, et al. "SBI--A toolkit for simulation-based inference." *arXiv preprint arXiv:2007.09114* (2020).
3. Radev, Stefan T., et al. "BayesFlow: Learning complex stochastic models with invertible neural networks." *IEEE transactions on neural networks and learning systems* 33.4 (2020): 1452-1466.
4. Papamakarios, George, David Sterratt, and Iain Murray. "Sequential neural likelihood: Fast likelihood-free inference with autoregressive flows." *The 22nd international conference on artificial intelligence and statistics*. PMLR, 2019.
5. Jan Boelts, Jan-Matthis Lueckmann, Richard Gao, Jakob H Macke (2022) Flexible and efficient simulation-based inference for models of decision-making eLife 11:e77220
6. Navarro, Daniel J., and Ian G. Fuss. "Fast and accurate calculations for first-passage times in Wiener diffusion models." *Journal of mathematical psychology* 53.4 (2009): 222-230.
7. Voss, Andreas, and Jochen Voss. "Fast-dm: A free program for efficient diffusion model analysis." *Behavior research methods* 39.4 (2007): 767-775.

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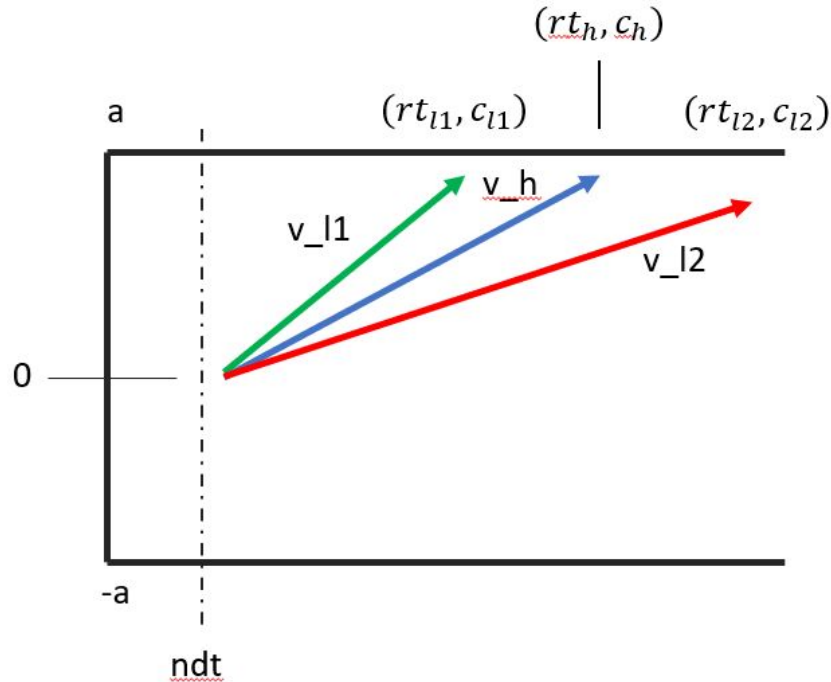
Observed Data Plot Bank



Seq 2 - Posterior predictive checks (RT vs. count)



Par2 Model



$$rt = \max(\underline{rt}_h, rt_{l2}, rt_{l1})$$

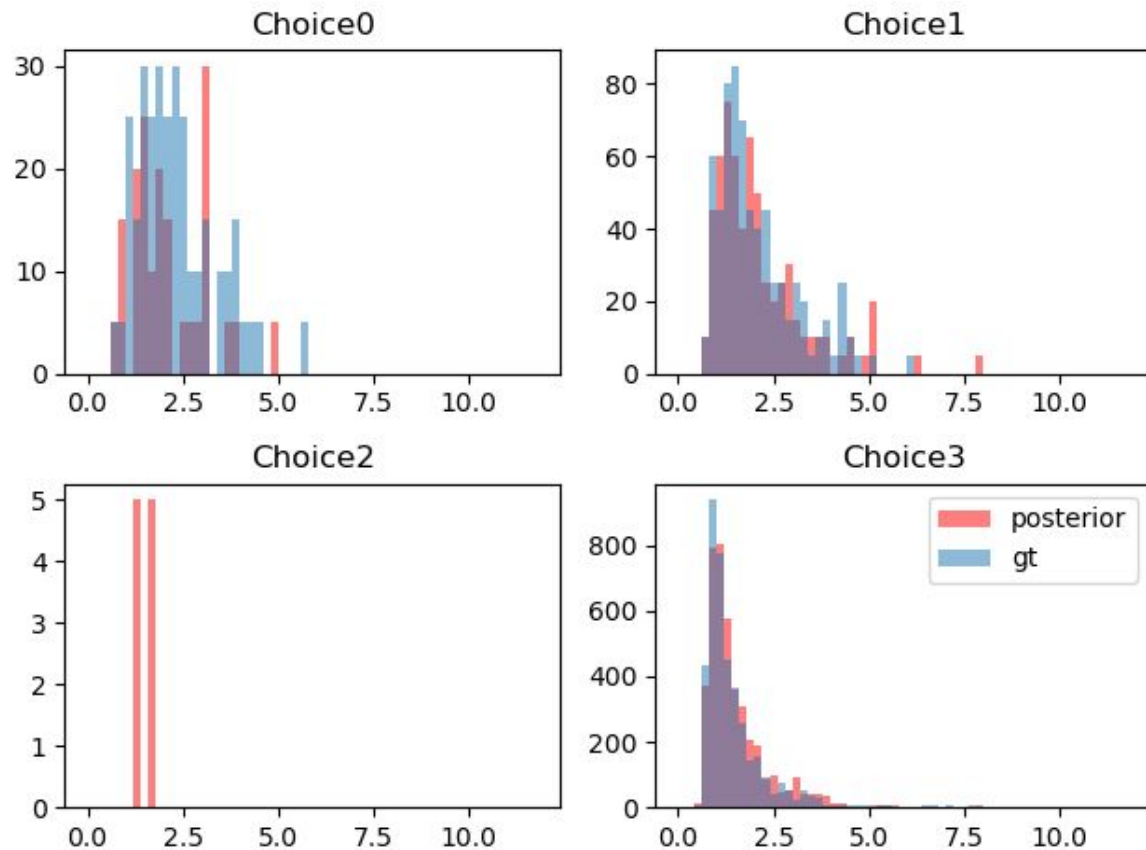
$$c = (\underline{c}_h, c_{l(c_h)})$$

$$\Delta x_h(t) = \Delta t * v_h + \sqrt{\Delta t} * N(0, 1)$$

$$\Delta x_l(t) = \Delta t * v_l + \sqrt{\Delta t} * N(0, 1)$$

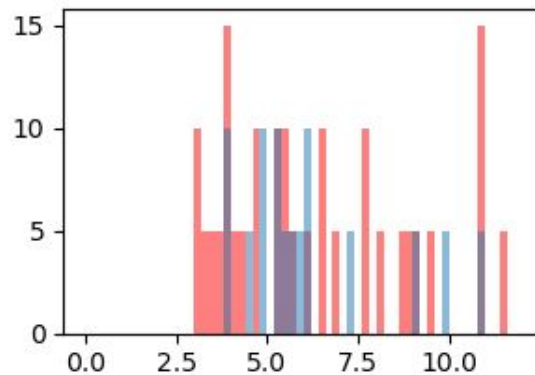
Parameters: $(v_h, v_{l1}, v_{l2}, a, \tau)$

Parallel - posterior predictive check

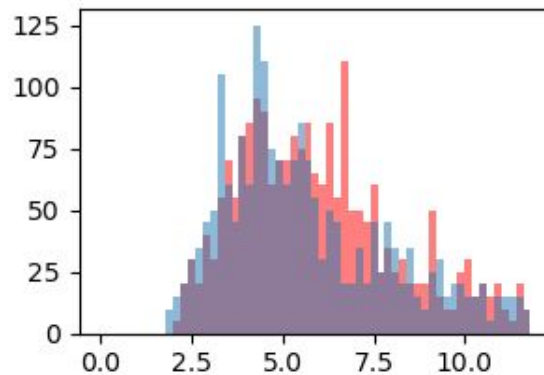


Leak - posterior predictive check

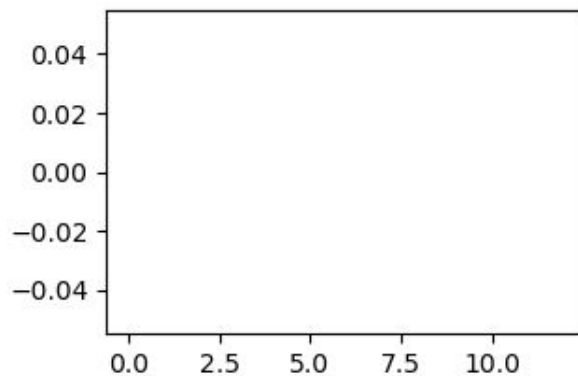
Choice0



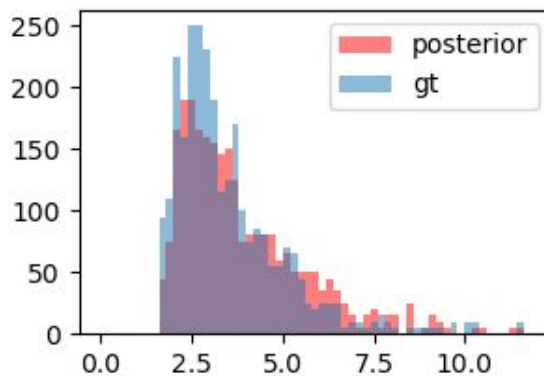
Choice1



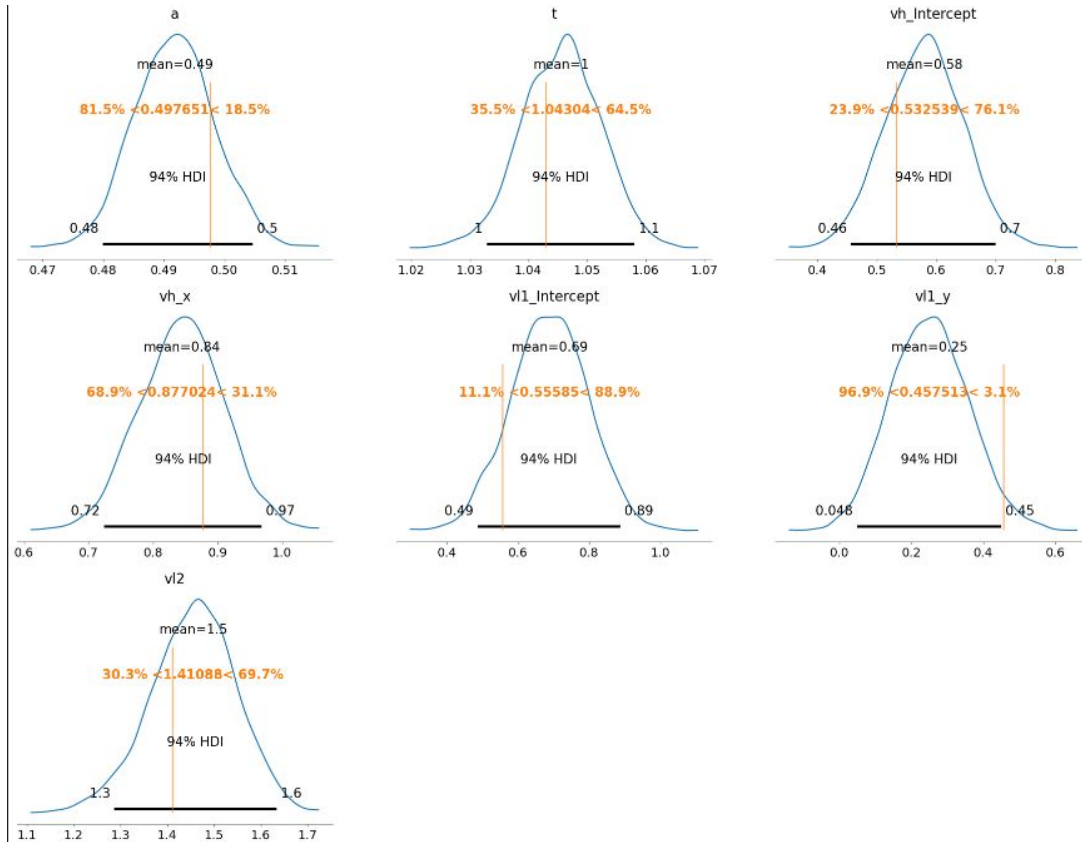
Choice2



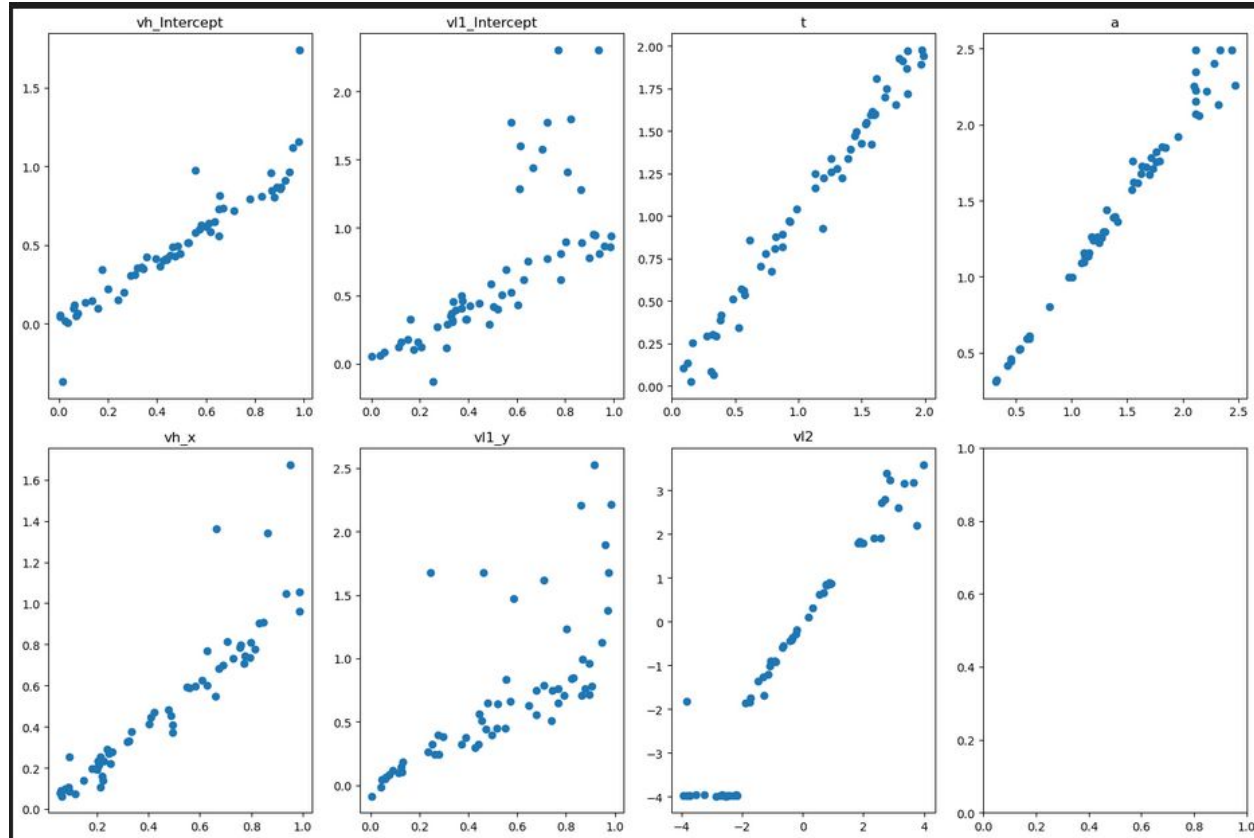
Choice3



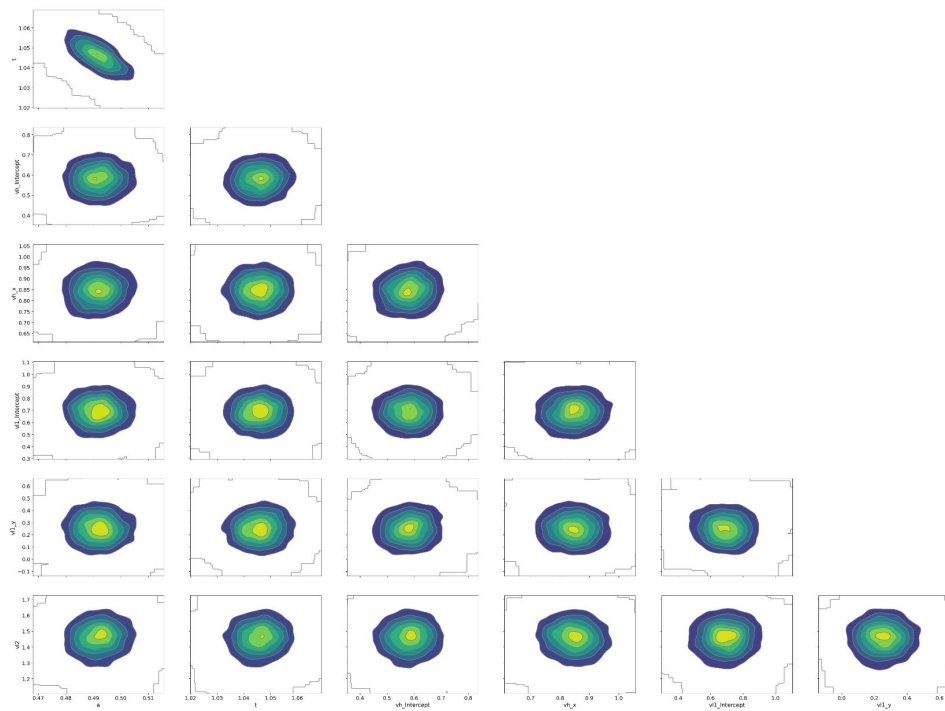
Single participant - synthetic data - sequential model



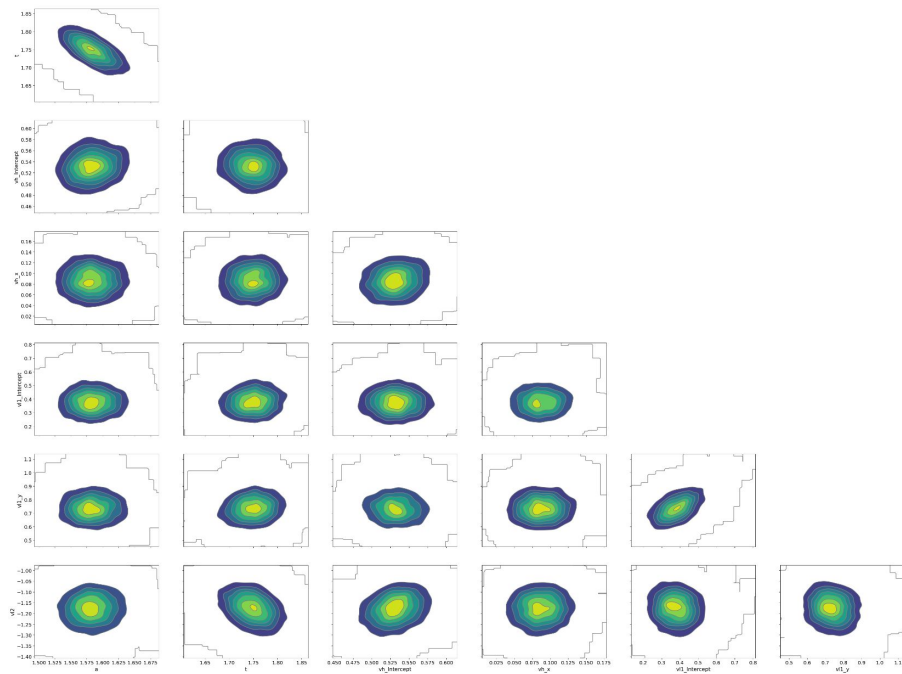
Parameter recovery - Sequential



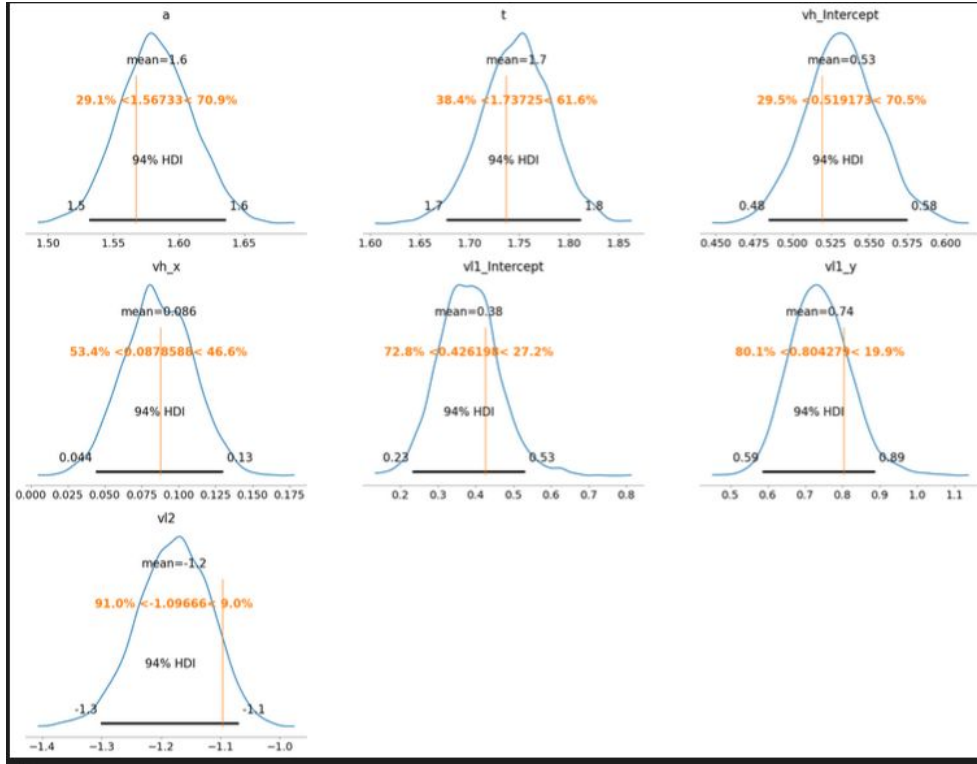
Single participant - empirical data

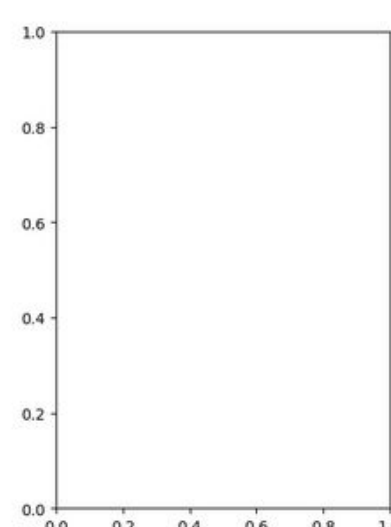
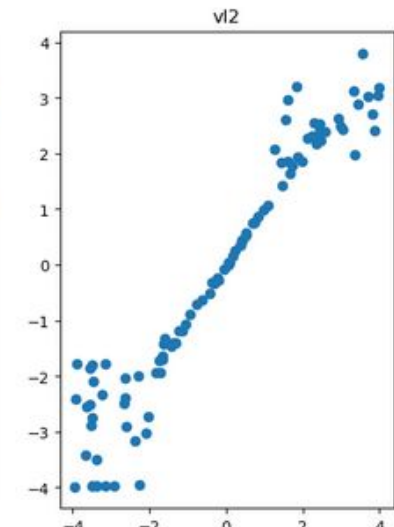
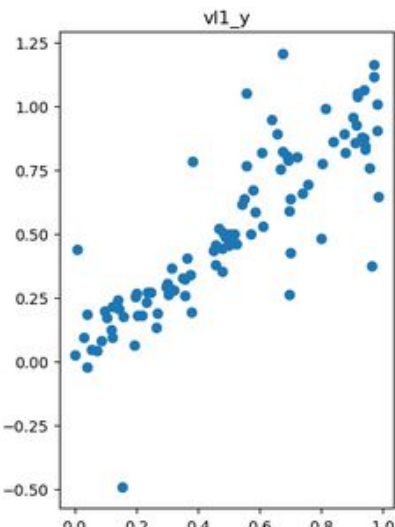
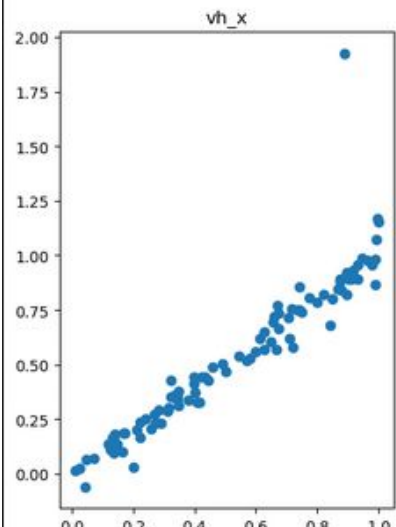
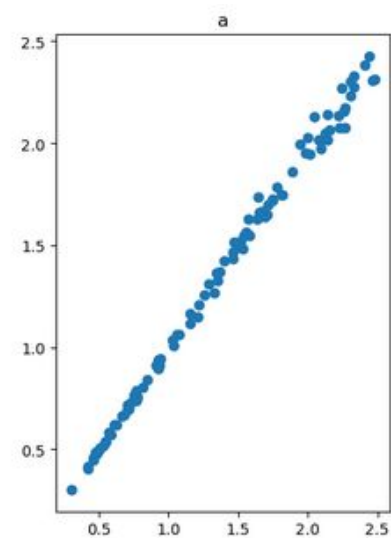
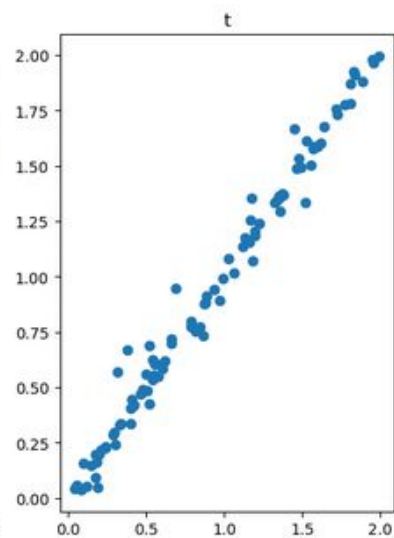
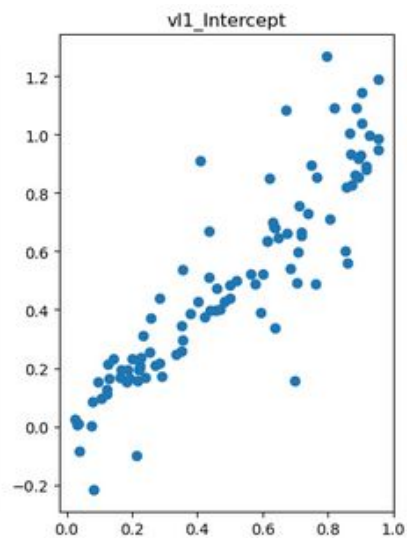
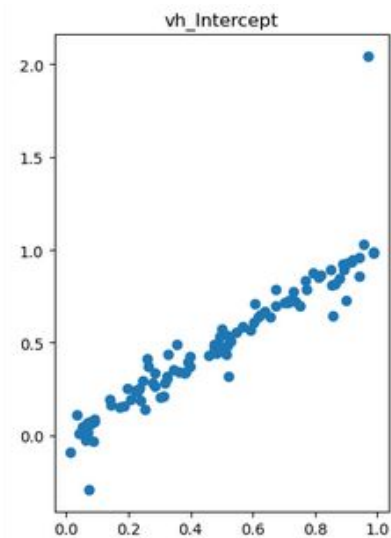


Single participant - empirical data

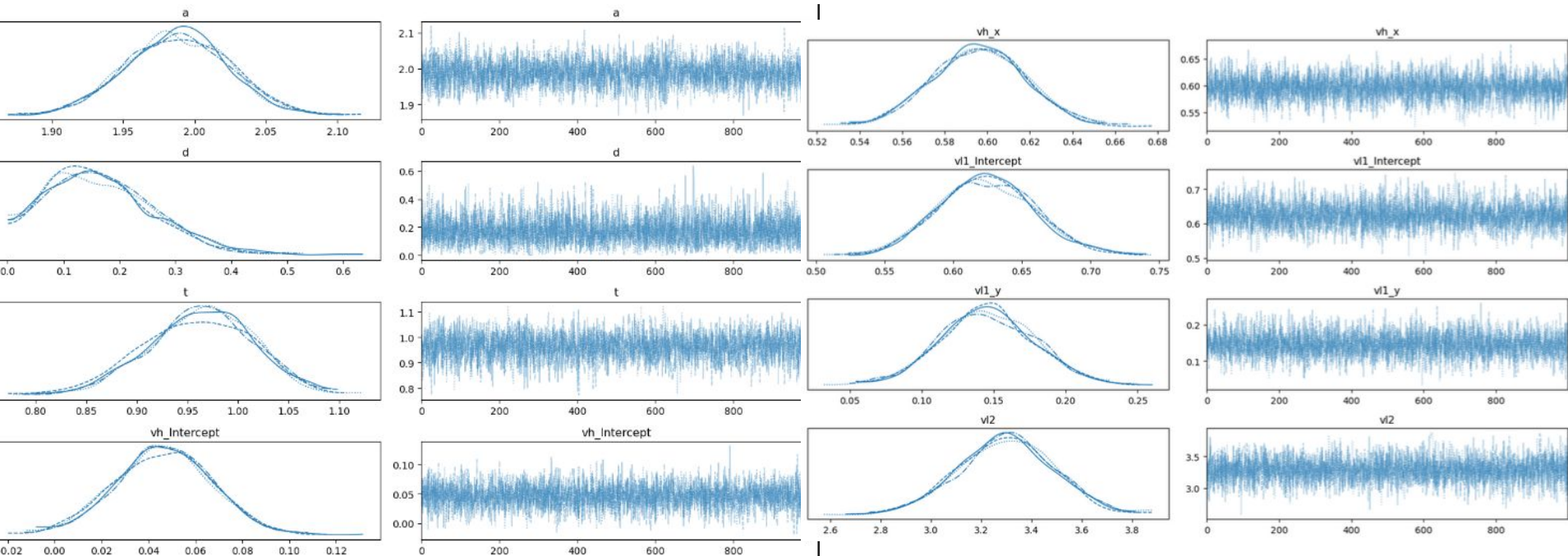


Single participant - synthetic data - parallel model

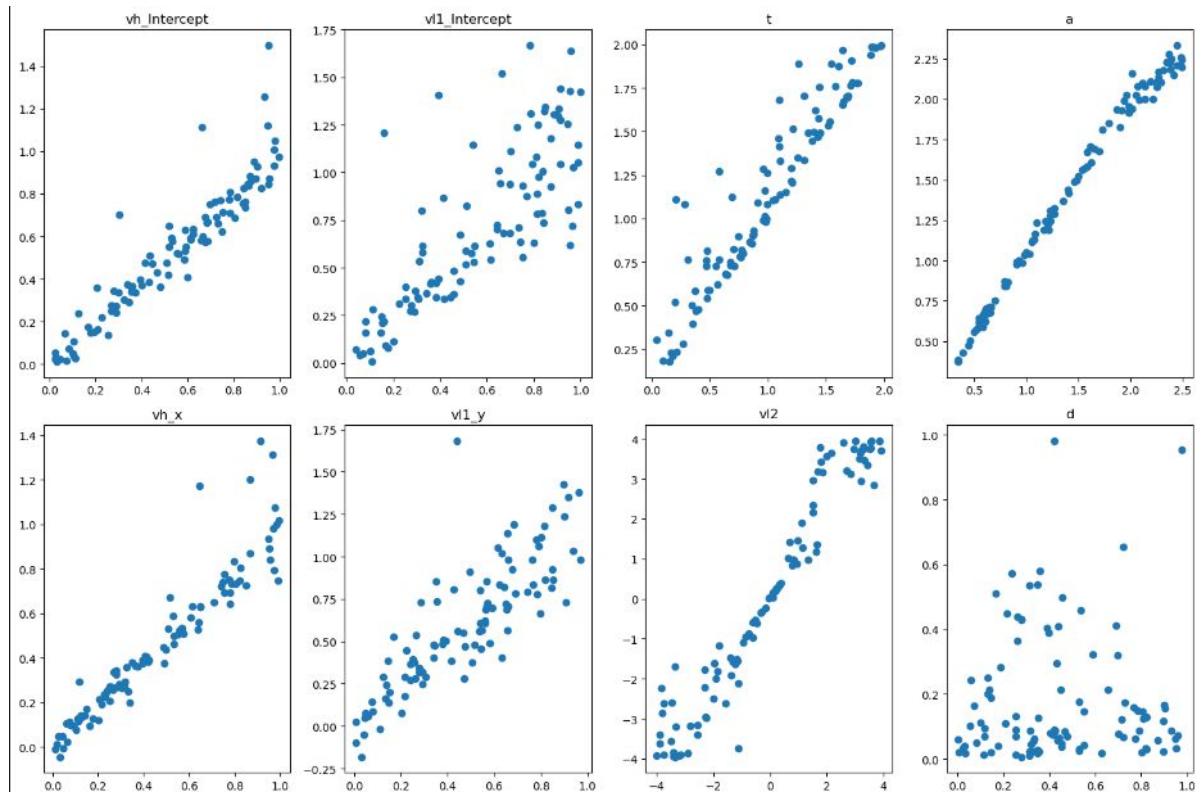




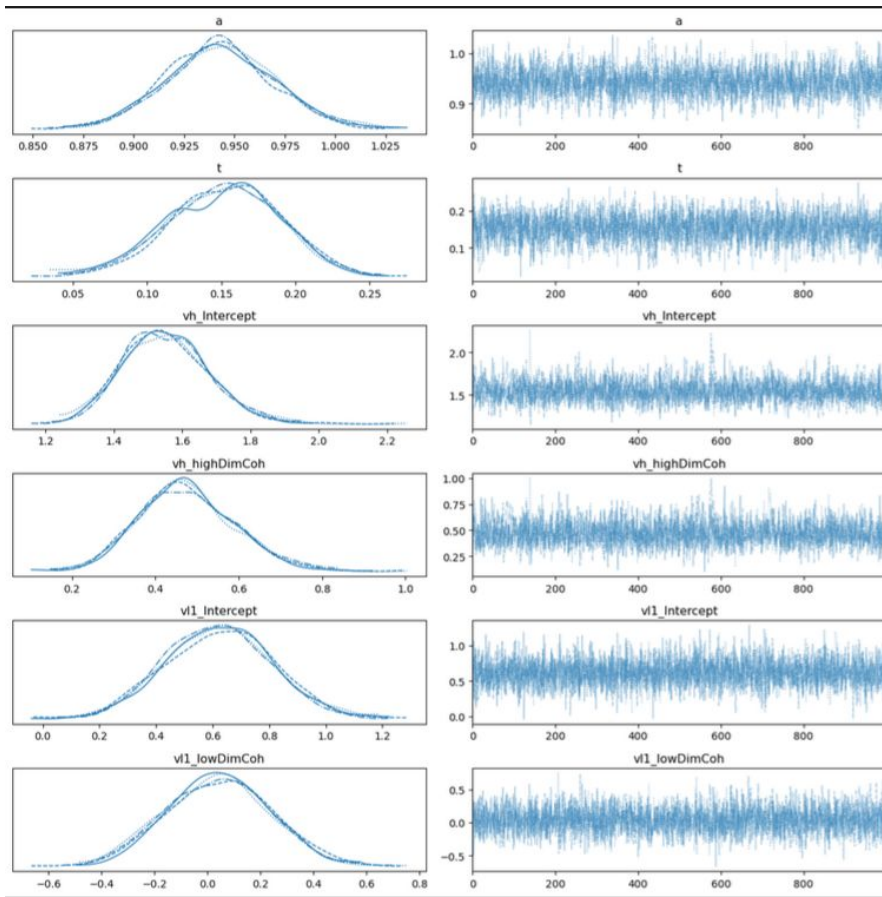
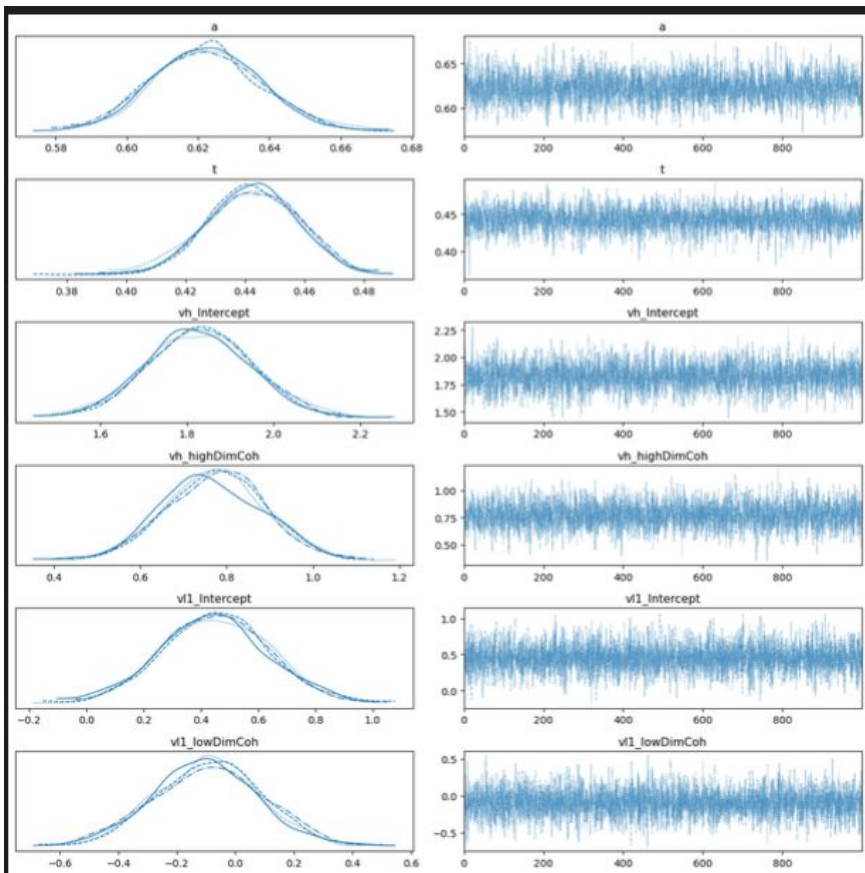
Single participant - synthetic data - leak model



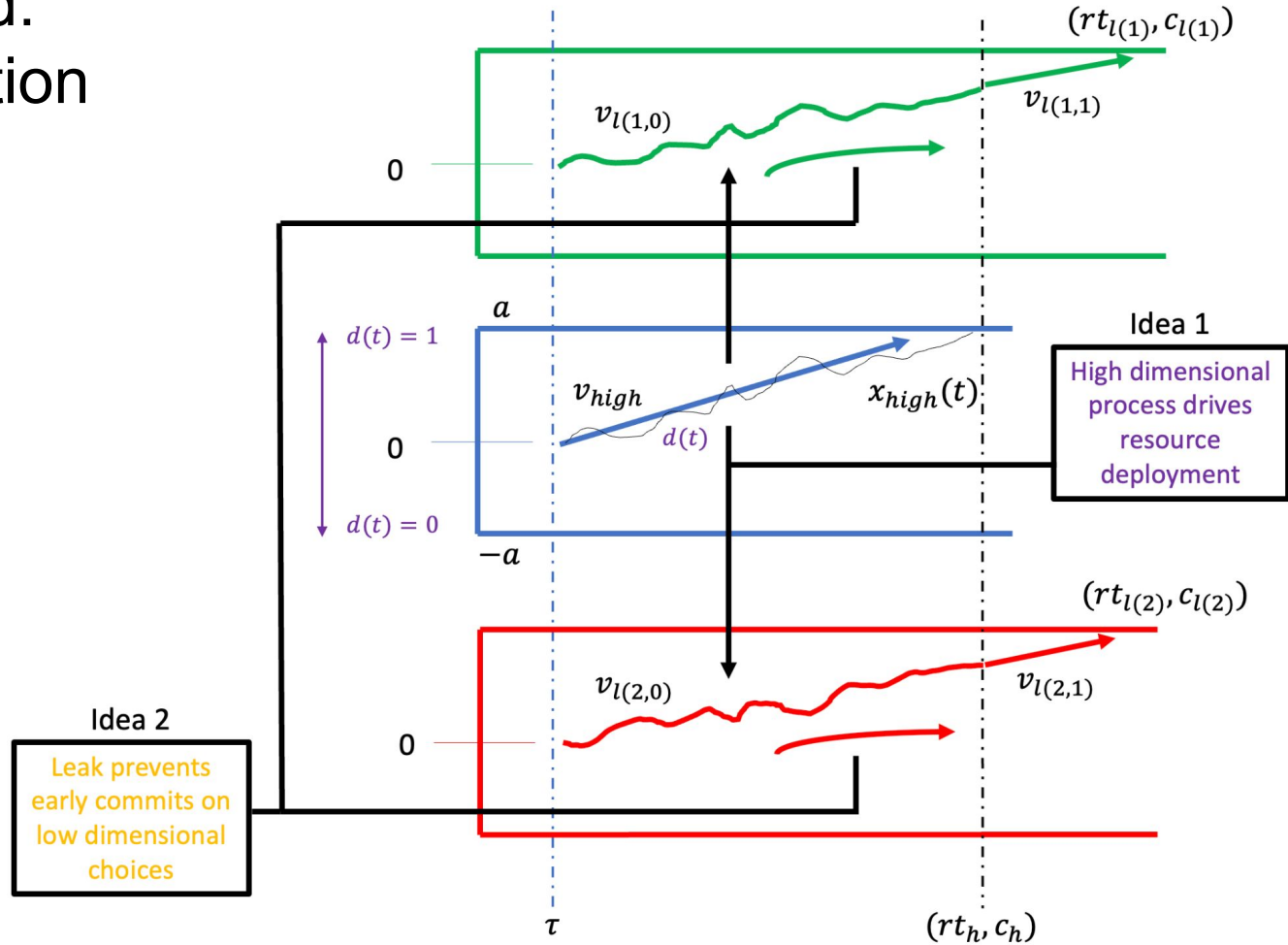
Parameter recovery - Leak



Empirical data - Sequential



Model 3... Hybrid: Resource Allocation



Model 3... Hybrid: Resource Allocation

$$\Delta x_{l(1,0)}(t) = \Delta t * v_{l1}(t) - (g * x_{l(1,0)}(t)) + \sqrt{\Delta t} * N(0,1)$$

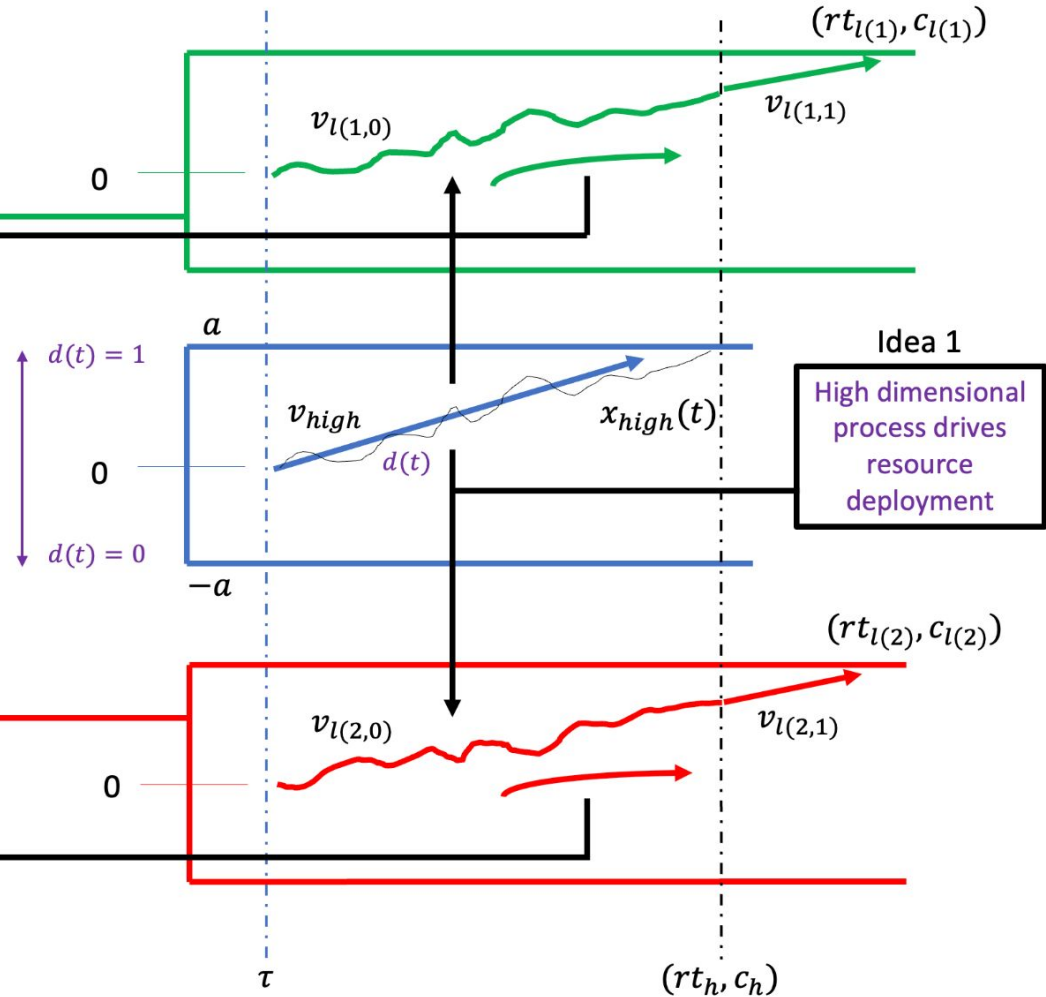
$$\Delta x_{l(1,1)}(t) = \Delta t * v_{l1}(t) + \sqrt{\Delta t} * N(0,1)$$

$$\Delta x_{l(2,0)}(t) = \Delta t * v_{l1}(t) - (g * x_{l(2,0)}(t)) + \sqrt{\Delta t} * N(0,1)$$

$$\Delta x_{l(2,1)}(t) = \Delta t * v_{l2}(t) + \sqrt{\Delta t} * N(0,1)$$

Idea 2

Leak prevents
early commits on
low dimensional
choices



Model 3... Hybrid: Resource Allocation

$$\Delta x_{l(1,0)}(t) = \Delta t * v_{l1}(t) - (g * x_{l(1,0)}(t)) + \sqrt{\Delta t} * N(0, 1)$$

$$\Delta x_{l(1,1)}(t) = \Delta t * v_{l1}(t) + \sqrt{\Delta t} * N(0, 1)$$

$$\Delta x_{l(2,0)}(t) = \Delta t * v_{l1}(t) - (g * x_{l(2,0)}(t)) + \sqrt{\Delta t} * N(0, 1)$$

$$\Delta x_{l(2,1)}(t) = \Delta t * v_{l2}(t) + \sqrt{\Delta t} * N(0, 1)$$

Idea 2

Leak prevents
early commits on
low dimensional
choices

